

The molecular theory of polyelectrolyte solutions with applications to the electrostatic properties of polynucleotides

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I. INTRODUCTION	181
II. THE MOLECULAR BASIS OF POLYELECTROLYTE BEHAVIOUR	182
A. Counterion condensation – a directly observable property of polyelectrolytes in solution	182
B. Nature of the condensed state; delocalized binding	185
C. Molecular theory of counterion condensation; the counterion diffusion potential	187
III. BINDING OF METAL IONS IN THE AQUEOUS DNA/ MCl_N SYSTEM	194
A. Charge fraction of DNA and mode of binding	194
B. The free energy of binding	195
C. Colligative properties	195
D. Transport properties	196
E. The dissociation-field effect	198
F. Specific ionic effects	199
IV. THEORY OF DELOCALIZED BINDING OF N -VALENT CATIONIC SUBSTANCES TO POLYNUCLEOTIDES IN AQUEOUS EXCESS 1 : 1 SALT	201
A. Introduction	201
B. One-variable theory	
1. Minimization of the free energy	201
2. Chemical model	208
C. Two-variable theory	211

- D. *Divalent metal ion binding* 212
 - 1. *Summary of results for $N = 2$* 212
 - 2. *Comparison of theory with Mg^{2+} binding data* 213
 - 3. *Conclusion* 214
- E. *Binding of small cations with complex structure* 215
 - 1. *Introduction* 215
 - 2. *Oligolysine* 215
 - 3. *Spermine* 217
 - 4. *Lys-trp-lys* 217
 - 5. *Discussion and conclusion* 217
- F. *Binding of high-molecular-weight cationic substances* 219
 - 1. *Introduction* 219
 - 2. *The polylysine/DNA complex* 220
 - 3. *The complex of lac repressor with non-operator DNA* 221
 - 4. *Repressor/operator complex* 221

V. IONIC EFFECTS ON THE DNA HELIX-COIL TRANSITION

223

- A. *Electrostatic free energy and entropy of transition* 223
- B. *Dependence of T_m on ionic strength* 223
- C. *Role of the counterion diffusion potential* 224
- D. *Effect of strong electric fields on T_m* 225
- E. *Effect of pH on T_m ; average axial phosphate spacing of single strand as DNA a function of pH* 226
- F. *Ionic effects of strandedness on T_m* 228
- G. *Effect of Mg^{2+} on T_m* 229
- H. *Ionic effects on the renaturation rate* 231
 - I. *The mistake of adopting the Debye-Hückel approximation* 232

VI. ELECTROSTATIC CONTRIBUTION TO DNA STIFFNESS 233

- A. *Harrington's measurements of persistence length* 233
- B. *Polyelectrolyte theory of the persistence length* 234
- C. *Discussion of the numerical results* 237
- D. *Implications for DNA condensation and nucleosome formation* 238
- E. *Further discussion* 238

I. INTRODUCTION

Although the importance of the polyelectrolyte character of DNA has been recognized for some time (Felsenfeld & Miles 1967), few of the implications have been explored, primarily because of a lag in translating the breakthroughs in polyelectrolyte theory of the last decade into a form that is well adapted to the analysis of the specialized problems of biophysical chemistry. Perhaps an analogous situation existed in the field of protein chemistry during the period after the formulation and confirmation of the Debye-Hückel theory of ionic solutions but before Scatchard's incorporation of the theory into his analysis of the binding properties of proteins. An achievement for polynucleotide solutions parallel to Scatchard's was recently presented by Record, Lohman, & de Haseth (1976) and further developed and reviewed by Record, Anderson & Lohman (1978). Record and his colleagues have succeeded in the self-consistent incorporation of the central result of the theory of polyelectrolyte solutions – counterion condensation – into the thermodynamic description of multiligand binding. A general conclusion of major importance arising in a systematic way from their analysis is that the release of counterions is a substantial contribution to the driving force for the binding of cationic substances to polynucleotides. A limitation of the thermodynamic approach, however, is a sharply reduced capacity for numerical prediction.

Indeed, polyelectrolyte theory has been a successful example of statistical mechanical methodology, whereby assumptions are made on the molecular level, a molecular model is constructed on the basis of these assumptions, and quantitative information about the behaviour of the model is then derived from well-known computational rules. In optimal cases – and polyelectrolytes present such a case – the parameters of the model are independently known, and then the results of the computation provide *a priori* numerical predictions that may be directly compared with the results of measurement. Good agreement strengthens the belief that the model is an accurate representation of reality; the experimentally observed behaviour has been successfully explained in terms of intermolecular forces and molecular movements. Systematic discrepancy between prediction and observation may also be of value, in so far as specific aspects of physical reality that may have been omitted from the model may be revealed. In this paper I have attempted to extend the theory of polyelectrolyte solutions to cases of

interest in polynucleotide chemistry. Its statistical mechanical character, and hence its full predictive and explanatory capability, is retained.

I shall begin with a detailed discussion of the ionic state of an extended polynucleotide structure in an aqueous environment of simple counterions, such as the alkali metal and alkaline earth cations. I shall then develop the concept that in such an environment the polynucleotide is 'potentiated' by its counterions for the spontaneous occurrence of any process that results in the relief of the electrostatic stress on the polynucleotide set up by the repulsive interactions among the phosphate groups. Binding of multivalent cationic substances is only one process of this type. Others include transitions to conformation of lower charge density – the DNA helix-coil transition being the prototypical example – and the tendency of DNA to straighten spontaneously from a locally bent configuration (a thermodynamic consequence of its very long persistence length).

To varying degrees of detail, but always on a molecular, as distinct from thermodynamic, level, I will be able to discuss the binding to polynucleotides of cationic substances as diverse as sodium and magnesium ions, spermine, polylysine, and *lac* repressor. At this stage perhaps it will be forgiven if the variation in mathematical detail is inversely related to the structural complexity of the cation. The analysis of ionic effects on conformational transitions includes those induced by strong electric fields as well as by temperature increases. I have also treated the local bending of DNA as a differential conformational transition to a state of higher charge density. The result suggests that DNA would not be a stiff molecule (on the scale of its persistence length) in the absence of the phosphate repulsions. An obvious corollary is that DNA will collapse, or condense, if sufficient charge neutralization by bound cationic substances occurs.

II. THE MOLECULAR BASIS OF POLYELECTROLYTE BEHAVIOUR

A. *Counterion condensation – a directly observable property of polyelectrolytes in solution*

Except for cases involving binding to the bases, the interaction of extended polynucleotide configurations with small ions is governed by the general laws of polyelectrolyte solutions. Indeed, the commonly heard dogma that a polyelectrolyte solution of low normality belongs

nevertheless, because of the high counterion concentration in the immediate vicinity of the polymeric chain, to the realm of concentrated salt solutions – about which little or nothing of a general nature can be quantified – has turned out not to be true. All particular effects in the locally concentrated region are subsumed by the general phenomenon of counterion condensation, which, together with ionic screening interactions at long range, familiar from the classical theory of dilute small-electrolyte solutions, provides a framework for polyelectrolyte solutions (Manning, 1969, 1972*a*, 1974, 1977*a*, 1978; Bailey, 1973; Iwasa, 1977) capable of predicting with remarkable accuracy a large variety of measurable properties. The concept of counterion condensation is a radical departure from the ideas that have traditionally dominated the biochemical literature in this field. When first introduced as the basis for polyelectrolyte theory (Manning, 1969), its status as a physically sharp phenomenon, as opposed to a convenient mathematical device, was uncertain. Experimental progress has been rapid, however, and has established condensation as a fundamental physical property of polyelectrolyte solutions. It has become easy, therefore, to avoid all possibility of investing the definition of counterion condensation with an abstract quality, simply by describing its measurement.

Consider therefore the following ultrasonic absorption experiment (Zana *et al.* 1971). NaOH is added to an aqueous solution of carboxymethylcellulose (CMC), initially in its completely protonated form. No differential ultrasonic absorption (with respect to titration with tetramethylammonium hydroxide) can be detected until the titration has proceeded to a critical extent of neutralization; whereupon, with continuing titration, absorption increases up to complete neutralization of the polyacid. Addition of more Na⁺ in the form of its chloride salt has no further effect on the ultrasonic absorption. The explanation is without ambiguity (Zana *et al.* 1971). In the initial stages of the titration the Na⁺ counterions are not in close interaction with the CMC polyion. At a critical degree of neutralization, however, further addition of OH⁻ results not only in chemical neutralization of carboxyl protons but also their replacement near the CMC chain by Na⁺ counterions added simultaneously. The volume fluctuations due to interpenetration of the electrostricted hydration layers surrounding both the Na⁺ ion and the CMC polyion are detected by the ultrasonic probe. Sodium ions continue to ‘condense’ on the polyacid as added until the titration end-point is reached. All carboxyl groups are now deprotonated, but

the charge fraction of the polymer – that is, the charge per carboxylate group – is less than unity if one counts the condensed Na^+ ions as part of the polymeric molecular unit. The charge fraction is then equal to α_{crit} , the critical degree of neutralization for the onset of differential ultrasonic absorption. Further addition of Na^+ as its chloride salt does not lead to increased Na^+ binding; the charge fraction is stable to increases of ionic strength. Conceptually similar experiments have been reviewed elsewhere (Manning, 1977*a*).

For the case of DNA volume effects are so small (an important point that will be expanded in subsequent discussion) that direct ‘tracking’ of the counterion itself appears to be necessary. Sodium-23 NMR line-width measurements have established that the charge fraction of DNA in an environment of excess Na^+ is invariant to variations of ionic strength at least up to 0.5 M (Anderson, Record & Hart, 1978). Under these conditions there remain many electrostatically uncompensated phosphate groups, as demonstrated by NMR-monitored titration of tetrabutylammonium DNA by Na^+ ions; association of Na^+ with the DNA polyion is indicated up to $75 \pm 10\%$ compensation of the phosphate charge but no further (Anderson *et al.* 1978). NMR charge-fraction experiments on a variety of other polyelectrolytes have been reviewed (Manning, 1977*a*).

Similar results are obtained for the interaction of Mg^{2+} with DNA. Dialysis-monitored titration of DNA by Mg^{2+} in excess 1 : 1 salt yielded about 85–86% maximum DNA charge neutralization by bound Mg^{2+} (Skerjanc & Strauss, 1968; Bhat, 1974). Over a limited range of added counterion, a minimum non-zero charge fraction independent of counterion concentration has also been confirmed by direct NMR measurement for the cobalt and manganese polyphosphate systems (Speg̃t & Weill, 1976).

Thus, as a matter of physical observation, polyelectrolytes in general, and DNA in particular, in an environment containing only one species of small cation like Na^+ and Mg^{2+} , possess a saturated charge fraction less than unity but well above zero. The data reveal a further, more restrictive generalization; in each case the value of the charge fraction depends only on the valence of the counterion and the *axial charge density* of the polyelectrolyte. The charge fractions of fully deprotonated sodium polyacrylate (carboxylate groups), sodium polystyrenesulphonate and sodium polyphosphate are equal (Manning, 1975; Kielman, van der Hoeven & Leyte, 1976). If b is the average axial charge spacing

(along the helical axis for DNA, along the contour axis for more flexible polymers) and if a dimensionless structural parameter ξ is defined as proportional to the charge density,

$$\xi = q^2/\epsilon kTb, \quad (1)$$

where q is the protonic charge, ϵ the bulk dielectric constant of solvent, k Boltzmann's constant, and T the Kelvin temperature, then a quantitative formulation of the physical observation is that the saturated charge fraction is close to $(N\xi)^{-1}$, where N is the valence (without sign) of the counterion. For water at 25 °C, $\xi = (7.1/b)$ if b is expressed in Å (the temperature dependence is small and may be neglected for most purposes). The B form of DNA has step height 3.4 Å between base pairs, i.e. between phosphate pairs, so its value of b is 1.7 Å and $\xi_{\text{DNA}} = 4.2$. Then the charge fractions of DNA in aqueous Na⁺ and Mg²⁺ environments are, respectively, 0.24 and 0.12.

The operational definition of counterion condensation can therefore be formulated thus: *the mode of binding of counterions of valence N to a polyelectrolyte is called condensation if, in an environment containing only the single counterion species, the charge fraction of the polyelectrolyte equals the constant value $(N\xi)^{-1}$ over a broad concentration range.*

B. Nature of the condensed state; delocalized binding

The extent to which condensed counterions are dehydrated and localized near discrete groups, that is, the 'tightness' of binding, may be expected to vary both with the nature of the charged groups and the counterion species (Strauss, 1974). At one extreme the counterion may be fully dehydrated and directly coordinated to several neighbouring charged groups. About half of the Mn²⁺ and Co²⁺ ions condensed on the polyphosphate chain are bound in this fashion (Spegt & Weill, 1976). At the other extreme the condensed counterion is said to be 'delocalized'. It retains the fully hydrated state characteristic of a pure aqueous environment while moving in an unrestricted and random way along the length of the polyion chain. If, for example, about half of the Mn²⁺ and Co²⁺ population condensed on polyphosphate is, as mentioned, completely dehydrated, the other half has been shown to retain at least a complete inner hydration layer (Spegt & Weill, 1976). NMR relaxation measurements have demonstrated the nearly unrestricted movement of possibly the entire population of Na⁺ and Rb⁺ condensed on DNA (Reuben, Shporer & Gabbay, 1975). The extremely small

specificities observed in differential dialysis experiments among alkali metal ions bound to DNA (J. Span & U. P. Strauss, unpublished) indicates the delocalized binding mode for condensed Li^+ and K^+ as well. Relaxation kinetic studies on the binding of Mg^{2+} and Ca^{2+} to poly A and poly C reveal diffusion control of the binding process, a result consistent with delocalized movement of these alkaline earth cations when bound and inconsistent with tight dehydrated binding to localized sites on the polynucleotide (Pörschke, 1976*a*). There is no reason to anticipate dissimilar behaviour for DNA, and the very small dilatometric effects exhibited by the binding of Mg^{2+} to DNA (in contrast to much larger effects for a number of synthetic polyelectrolytes) is indeed direct evidence that Mg^{2+} binds to DNA with almost no dehydration (Skerjanc & Strauss, 1968).

That alkali metal and alkaline earth cations condensed on extended polynucleotide conformations are almost fully hydrated and delocalized on the polyion surface is in fact their *expected* behaviour. Each fixed phosphate group is monovalent and rigidly held (Olson, 1975) at a great distance, about 7 Å, from its nearest-neighbour phosphate groups. Singly charged phosphate monomers do not interact strongly with cations. Activity coefficients of aqueous MH_2PO_4 solutions decrease (i.e. the ionic interactions become stronger) in the order of decreasing *hydrated* radius of the alkali metal ion M^+ (Li^+ interacts more weakly than K^+) (Robinson & Stokes, 1959). No ion-pair formation between Mg^{2+} and H_2PO_4^- , monovalent glucose 1-phosphate, or monovalent glycerol 2-phosphate could be detected by sensitive electrochemical measurements (Greenwald, Redish & Kibrick, 1940; Clarke, Cusworth & Datta, 1954). It should be realized that, even if ion-pairing with the monomer is present to a slight extent, delocalization on the polymer would still be likely to occur because of the long-range overlap of coulomb fields. Tight localized binding (site binding) is expected to occur for a polyelectrolyte with monovalent charged groups only if the chain is sufficiently flexible or the nearest-neighbour charge spacing is sufficiently close that two or more groups can co-operate in holding the cation, i.e. when the 'monomer' is effectively multivalent (Strauss & Siegel, 1963). Extended polynucleotide conformations do not satisfy this requirement (Skerjanc & Strauss, 1968; Olson, 1975). Significant binding of alkali metal or alkaline earth cations to the bases in aqueous solution has been ruled out by differential dialysis studies (Shapiro, Stannard & Felsenfeld, 1969*b*).

The qualitative difference, *as a matter of physical observation*, between polyelectrolyte-counterion binding and association reactions of oppositely charged small ions cannot be overemphasized. The latter is governed by mass action. The *extent* of ion pairing and the *tightness* of the complex are inseparable; the greater the association constant, the greater will be both the extent of reaction (number of ion pairs) and the tightness (stability) of each pair. For a polyelectrolyte in an environment of a single counterion species, however, the extent and tightness of binding are uncorrelated. Polyphosphate and DNA, which possess, roughly, about the same axial charge densities ($\xi_{PP} = 2.8$, $\xi_{DNA} = 4.2$), bind about the same number of Na^+ ions in accordance with the condensation principle; but condensed Na^+ ions are bound loosely to DNA with little or no dehydration and localization, while Na^+ ions condensed on polyphosphate are extensively dehydrated (Strauss, 1974), although probably still not strongly localized (Kielman *et al.* 1976). In accordance with its higher valence, more Mg^{2+} than Na^+ is bound on an equivalent basis to the synthetic polyelectrolyte PVS (with weakly binding sulpho-nate groups that are comparable to the monovalent phosphates on polynucleotides), but Na^+ is actually the more *tightly* bound of the two if extent of dehydration per equivalent is used as the criterion (Hen & Strauss, 1974).

Of course if the affinity of a counterionic species for sites on the polyion contains an appreciable covalent, or other non-ionic, component, then the situation is different. The binding of protons to nucleotide bases or to carboxylate groups on polyacrylate, for example, does not conform to the condensation requirement, but rather obeys a mass-action law with an association constant dependent on the extent of binding. There is also a formal similarity to a mass-action binding distribution when a multivalent species (Mg^{2+} or small oligolysines, for example) equilibrates by purely electrostatic interaction with DNA in the presence of excess univalent salt, although important qualitative differences remain. This situation, which is of the greatest practical interest for the biologist, will be discussed subsequently.

C. *Molecular theory of counterion condensation; the counterion diffusion potential*

It has been proved (Manning, 1977*a*, 1978) that condensation of counterions is a simple consequence of the spatial correlation of fixed charge centres along a single axis characteristic of all linear polyelectro-

lytes (even flexible chains can be expected to be fairly 'straight' over distances of the order of a Debye length because of electrostatic repulsion among the fixed charges). Thus, condensation is a manifestation of long-range electrostatic forces, and its general occurrence regardless of the specific nature of the polyelectrolyte charges is thereby explained.

To investigate the consequences of long-range overlap of the spherical coulomb fields generated by each charged group (assumed univalent) on the polyion, the actual spatial distribution of groups with average axial spacing b may be replaced by a regular linear array of fixed univalent point charges of magnitude q and spacing b (Manning & Zimm, 1965; Manning, 1977*a*). Let θ_N be the number of associated counterions per fixed charge, with the understanding that all counterions M^{N+} in the system have the same valence N . Then the *effective* charge q_{net} on each element of the linear array is lowered from q by the factor $1 - N\theta_N$,

$$q_{\text{net}}/q = 1 - N\theta_N. \quad (2)$$

Each pair of effective charges is then assumed to contribute a repulsive term to the overall free energy of the form

$$(q_{\text{net}}^2/\epsilon) (|i-j|b)^{-1} \exp(-\kappa|i-j|b), \quad (3)$$

a Debye-Hückel screened coulomb potential between the i th and j th charges in the array, $|i-j|b$ being their fixed distance of separation. The Debye screening parameter κ is proportional to the square root of ionic strength (if simple electrolyte is present in excess over polyelectrolyte):

$$\kappa^2 = (8\pi) \times 10^{-3} L_{\text{AV}} (q^2/\epsilon kT) I, \quad (4)$$

where L_{AV} is Avogadro's number and I is the ionic strength. For excess 1:1 salt in water at 25 °C,

$$\kappa = (3.29 \times 10^7) c_1^{1/2} [\text{cm}^{-1}], \quad (5)$$

where c_1 is the molarity of excess univalent counterion (ionic strength of the salt). The theoretical extent of association θ_N will therefore be dictated by the deviation of the actual electrostatic free energy from the Debye-Hückel approximation to it, or, again, the departure from diffuse Debye-Hückel screening interactions (ion atmosphere) is theoretically attributed entirely to counterion association. Thus determined, the theoretical extent of counterion association satisfies the requirements for counterion condensation (see Section IIA).

The total free energy of the system is the sum of contributions G_i from various sources, of which screened repulsions of the form of equation

(3) are one. In general I shall define

$$\bar{g}_i = G_i/n_P RT, \quad (6)$$

where n_P is the total number of moles of charged groups on the polyions (number of moles of phosphate groups in a polynucleotide system) and R is the gas constant. I shall also assume that

$$I \gg c_P, \quad (7)$$

where c_P is the polyion equivalent concentration (molarity of phosphate groups for a polynucleotide), so that equations (4) and (5) may be used.

The work of charging the phosphate groups against their mutual electrostatic repulsion is obtained by summing equation (3) over all pairs (i, j) of effective charges in the linear array,

$$\bar{g}_{el} = -(\tau - N\theta_N)^2 \xi \ln(\tau - e^{-\kappa b}), \quad (8)$$

where ξ has been defined by equation (1). Note that the summation, the mathematical device for making the 'overlap' of coulomb fields a quantitative concept, has drastically transformed the qualitative nature of the individual spherical screened coulomb fields; the exponential screening, equation (3), has become logarithmic.

Other contributions to the free energy that depend on θ_N are given by the mixing of free M^{N+} cations, bound M^{N+} cations, and solvent molecules. Lumped together, these terms can be written as

$$\bar{g}_{mix} = \theta_N \ln(10^3 \theta_N V_P^{-1}/c_N), \quad (9)$$

where terms independent of θ_N have been omitted. The argument of the logarithm in equation (9) is the ratio of the local concentration of bound M^{N+} cations to their molarity c_N in free solution. The quantity V_P is the volume of the region surrounding the polynucleotide within which M^{N+} cations are said to be 'bound'. The units of V_P are $\text{cm}^3/\text{mole } P$, so that the local concentration $10^3 \theta_N V_P^{-1}$ is expressed in molarity units, and the argument of the logarithm is dimensionless. The assertion that there are no other θ_N -dependent contributions to the free energy is equivalent to the assumption that the interaction of bound and free M^{N+} ions with the solvent is the same, and that the 'bound' M^{N+} ions may translate freely within the volume V_P centred along the long axis of the polynucleotide. This assumption is the natural way to incorporate the discussion in IIB into a model.

Turning to the qualitative features of the dependence of $\bar{g}_{el}$ and

$\bar{g}_{m1x}$ on θ_N , one sees that $\bar{g}_{el}$ is positive and decreases as θ_N increases, the expected behaviour for the contribution of phosphate-phosphate repulsions, which decrease in magnitude as more counterions bind and effectively lower the phosphate charge. This term alone, then, would lead to complete binding, $N\theta_N = 1$, or minimum electrostatic work of charging the polynucleotide, as the equilibrium state.

However, increased binding of M^{N+} cations lowers the entropy of the system, as expressed by $\bar{g}_{m1x}$, which increases as θ_N increases if, as will be seen to be true for cases of interest here, the local concentration of bound M^{N+} is greater than the concentration of free M^{N+} ,

$$10^3\theta_N V_P^{-1} > c_N. \quad (10)$$

By itself, $\bar{g}_{m1x}$ would lead to $\theta_N = 0$ as the equilibrium state; that is, complete dissociation. The actual equilibrium state, determined from minimization of the sum $\bar{g}_{el} + \bar{g}_{m1x}$, is a value of $N\theta_N$ intermediate between 0 and 1, a state of incomplete dissociation and incomplete binding.

The discussion thus far has held no surprises; *any* system of interacting molecules possessing binding capacity behaves similarly, the equilibrium state of incomplete binding representing the balance between maximization of entropy by dissociation and minimization of energy by binding. Polyelectrolyte binding is qualitatively distinctive, however. For a typical binding system dissociation tends to completion as the concentration tends to zero, because the entropy increase becomes infinite while the energy increase is essentially independent of concentration and remains finite. But for the polyelectrolyte case it can be verified from equations (8) and (4) that $\bar{g}_{el}$ increases without bound as c_N tends to zero. Moreover, the concentration variation is logarithmic, just as for $\bar{g}_{m1x}$. The consequence is that θ_N does not tend to zero with c_N but remains positive. The implications of this distinctive behaviour are far-reaching. (It should be noted that the validity of equation (8) requires neglect of end-effects, an excellent approximation for ionic strengths of practical interest and *polynucleotides*, as opposed to *oligonucleotides*.)

To proceed quantitatively, the operation of minimization of $\bar{g}_{el} + \bar{g}_{m1x}$ with respect to θ_N is performed, with the result that θ_N satisfies the following transcendental relation at the free energy minimum,

$$1 + \ln(10^3\theta_N V_P^{-1}/c_N) = -2N\xi(1 - N\theta_N) \ln(1 - e^{-\kappa b}). \quad (11)$$

In the limit $c_N \rightarrow 0$ the only value of θ_N that satisfies the equation is

$$N\theta_N = 1 - (N\xi)^{-1} \quad (12)$$

because only for this value will the right-hand side produce a term $-\ln c_N$ to cancel exactly the singular $-\ln c_N$ term on the left-hand side (see equation (4)). The value of V_P that corresponds to equation (12) in the low-concentration limit is

$$V_P = 4\pi e L_{\Delta V} N' \nu^{-1} (\nu + \nu') (\xi - N^{-1}) b^3, \quad (13)$$

where e is the base of natural logarithms, ν and ν' are the number of counterions and coions, respectively, in the formula for the small electrolyte that is present in excess and hence determines the ionic strength, and N and N' are the valences (without sign) of the counterion and coion, respectively. Note that the only polyelectrolyte parameter on which V_P depends is the axial charge spacing b . Equation (13) is obtained by substituting equation (12) into (11) along with (4) for κ , while linearizing the exponential in accord with the stipulation $c_N \rightarrow 0$. For a 1:1 salt

$$\theta_1 = 1 - \xi^{-1} \quad (14)$$

and

$$V_P = 41 \cdot 1 (\xi - 1) b^3, \quad (15)$$

where V_P has the units $\text{cm}^3/\text{mole P}$ if b is expressed in ångströms. Clearly, equation (12) and (13) are physically meaningful only for polyelectrolytes of structures such that $\xi \geq N^{-1}$, a condition satisfied by all known polynucleotide conformations. Thus, a polyelectrolyte molecule with $\xi > 1$, immersed in 1:1 salt of vanishingly small bulk concentration c_1 , is nevertheless surrounded by a local concentration c_1^{loc} of counterions

$$c_1^{\text{loc}} = 10^3 \theta_1 V_P^{-1} = 24 \cdot 3 (\xi b^3)^{-1}, \quad (16)$$

where c_1^{loc} has molarity units if b is expressed in ångströms. Table 1 lists these local concentrations of bound univalent counterions for several polynucleotide structures. Also listed in Table 1 are the values Δx of the thickness of cylindrical shells of volume V_P about the polynucleotide; to calculate Δx it was assumed that the double- and triple-stranded structures are cylinders of radius 10 Å, while the single-stranded structures all have 5 Å radii. As an example from Table 1 the charge fraction of native DNA in excess 1:1 salt is 0.24 (76% of the phosphate charge is compensated by bound M^+), and the bound cations

TABLE 1. *Parameters for bound univalent cations in the limit $c_1 \rightarrow 0$*

Polynucleotide	b (Å)†	ξ	θ_1	V_P (cm ³ /mole P)	Δx (Å)	c_1^{100} (M)
Poly A·2 poly U	1.0	6.9	0.86	252	5	3.52
Poly A·poly U	1.6	4.5	0.78	589	7	1.32
Native DNA	1.7	4.2	0.76	646	7	1.18
Poly A	3.1	2.3	0.57	1592	12	0.35
Single-stranded DNA	4.0	1.8	0.44	2104	12	0.21
Poly U	4.5	1.6	0.38	2247	12	0.17

† Record, Woodbury & Lohman (1976), pH 7.

translate freely within about 7 Å of the DNA surface (of course, there is rapid exchange between ions in the bound region and bulk solution), where their local concentration is about 1.2 M, even though their bulk concentration c_1 may approach zero. *The local concentration is not governed by the law of mass action; it does not tend to zero with the bulk concentration c_1 .*

It is now natural to wonder whether an approximate ionic strength c_1 can be specified below which, for practical purposes, the system has reached the mathematical limit $c_1 \rightarrow 0$. That is, over what range of ionic strengths c_1 are the parameters in Table 1 practically applicable? It may be verified that if the condition $\kappa b \ll 1$ is fulfilled, so that the exponential term on the right hand side of equation (11) can be linearized, then equations (14) and (15) for θ_1 and V_P satisfy the free-energy minimization condition equation (11). Thus, for concentrations c_1 such that the Debye screening length κ^{-1} is appreciably larger than the axial phosphate spacing b , Table 1 is applicable. For double- and triple-stranded structures, this requirement is not very restrictive, since the corresponding values of b are so small. For example, κ^{-1} equals 3.0 Å at 1 M according to equation (5), which may be compared to the value $b = 1.7$ Å for native DNA. It is expected, then, that Table 1 is applicable to native DNA for ionic strengths in 1:1 salt approaching 1 M. For the single-stranded structures anticipation of the applicability of Table 1 would be restricted to concentrations below about 0.1 M, for which $\kappa^{-1} = 9.6$ Å.

An equivalent and most appealing criterion for the applicability of Table 1 is $c_1 \ll c_1^{100}$. It may in fact be verified from equations (5) and (16) that $c_1/c_1^{100} = 2.7\kappa^2 b^2$ (use of the general expressions rather than

TABLE 2. *Number of bound counterions per phosphate (θ_1) as a function of ionic strength*

$\xi = 4.2$		$\xi = 1.5$	
c_1 (M)	θ_1	c_1 (M)	θ_1
0.0001	0.76	0.0001	0.33
0.001	0.76	0.001	0.33
0.01	0.76	0.01	0.35
0.1	0.77	0.05	0.39
0.5	0.80	0.10	0.42
1.0	0.83		

the numerical ones identifies the factor 2.7 as e , the base of natural logarithms), so that the conditions $c_1 \ll c_1^{\text{loc}}$ and $\kappa b \ll 1$ are indeed equivalent. Thus, Table 1 applies for ionic strengths approaching the listed values of the local concentrations of bound counterions.

In Table 2 these qualitative remarks are validated. Equation (15) for V_P is used in (11) with $N = 1$ and θ_1 is then evaluated by numerical iteration as a function of c_1 . For $\xi = 4.2$ (native DNA) Table 1 indicates that c_1^{loc} equals 1.2 M, while Table 2 shows that θ_1 retains its limiting value 0.76 up to $c_1 = 0.1$ M and does not deviate much from this value even at $c_1 = 1$ M. For $\xi = 1.5$ ($\sim$ poly U) Table 1 lists c_1^{loc} as about 0.2 M, and Table 2 shows that θ_1 remains at its zero-concentration limit up to $c_1 = 0.01$ M and is not immoderately different for concentrations five times higher.

The significance of the trend in θ_1 as a function of c_1 exhibited in Table 2 is obscure at the present time. The theory contains inconsistencies that become significant at the same concentrations at which θ_1 begins to deviate from its limiting value (Manning, 1977a). From the experimental side no variations of polyelectrolyte charge fractions with ionic strength have been reported; the charge fraction of DNA in NaCl is invariant up to 0.5 M, the highest ionic strength investigated by Anderson *et al.* (1978). The point to emphasize from Table 2 at present is the theoretical *invariance* of θ_1 up to ionic strengths c_1 approaching c_1^{loc} , not the variation of θ_1 at higher ionic strengths.

I can now formulate an evocative description of the ionic state of an extended polynucleotide structure in a solution environment of excess 1:1 salt of concentration (ionic strength) $c_1 \ll c_1^{\text{loc}}$. The structure is 'potentiated', in the sense that a large entropy increase $nR \ln (c_1^{\text{loc}}/c_1)$

would accompany the diffusion of n M^+ cations from the bound region of locally high concentration to the lower concentration in free solution. This statement is implicit in equation (9), since $c_1^{lo} = 10^3 \theta_1 V_P^{-1}$. The process of local diffusion of M^+ into bulk solution does not occur spontaneously only because its presence at high concentration near the polynucleotide is required to offset the repulsive electrostatic free energy of interaction among the phosphate groups, as described by equation (8). However, if the system can avail itself of another mechanism for relieving the stress of phosphate-phosphate repulsion, then M^+ will diffuse spontaneously into free solution as the alternate mechanism is implemented. Conversely and equivalently, the counterion diffusion potential $RT \ln (c_1/c_1^{lo})$ constitutes a thermodynamic driving force for the spontaneous implementation of the alternate mechanism of phosphate charge compensation. As the ionic strength c_1 approaches c_1^{lo} , or 1.2 M for DNA, the polyelectrolyte is no longer potentiated in this sense, and, to the extent that a given process consists of the relief of long-range electrostatic stress, that process will have lost its ability to occur spontaneously.

III. BINDING OF METAL IONS IN THE AQUEOUS DNA/ MCl_N SYSTEM

A. Charge fraction of DNA and mode of binding

This subject has already been covered in Section II. The theoretical charge fraction r of DNA in excess MCl_N (fraction of phosphate charge uncompensated by bound M^{N+}) is given by (12):

$$r = 1 - N\theta_N = (N\xi)^{-1} \quad (17)$$

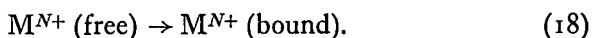
or 0.24 and 0.12 for $N = 1$ and $N = 2$, respectively. These values have been closely confirmed by direct measurement for Na^+ and Mg^{2+} in DNA solutions over wide ranges of ionic strength, and (17) has also been confirmed for Mn^{2+} and Co^{2+} in polyphosphate solutions; see Section II for references. There exists no reason at the present time to doubt the essential validity of (17) for all small univalent and divalent cations except those that may interact strongly with the nucleotide bases in native DNA, like Ag^+ and Cu^{2+} .

Good experimental evidence exists that Na^+ , Rb^+ , Mg^{2+} and Ca^{2+} are bound loosely to DNA, to the molecule as a whole rather than to discrete sites on it, and are in a state of essentially complete hydration

and free translational and rotational mobility (see Section II for references). This information has been incorporated into the theoretical model leading to the successful equation (17). The similarity in behaviour among all the alkali metal ions, as well as the nearly identical binding constants to native DNA of Mg^{2+} , Ca^{2+} , Mn^{2+} and Zn^{2+} (Daune, 1970; Clement, Sturm & Daune, 1973) suggests that *all* small cations, except those that can actually penetrate the native DNA structure and enter into strong interaction with the bases, are more or less associated with DNA in this delocalized mode. The burden of proof has now clearly shifted to those who would support a contrary position. As I shall point out in Section III F, the undisputed existence of small specific ionic effects does not constitute such proof.

B. The free energy of binding

Binding of M^{N+} cations to DNA in the system $\text{DNA}/\text{MCl}_N/\text{H}_2\text{O}$ can be formulated as a chemical reaction (Manning, 1978),



Note carefully that this reaction does *not* imply that free M^{N+} combines with a 'site' to form a cation-site complex; it states simply that M^{N+} is transferred from free solution into the spatial region V_P , where it is defined as bound. Application of the standard thermodynamic treatment of chemical reactions to equation (18) is fully equivalent to the treatment in IIC and yields the same results as (12) and (13). The integral free energy of reaction (18) – that is, the free energy when DNA is combined with its equilibrium number of counterions (and hence has charge fraction r given by (17)), relative to the non-equilibrium state of complete dissociation ($r = 1$) – is $\Delta\bar{g} = -5.3$ for $N = 1$ at ionic strength 0.1 M. That is, condensation of Na^+ ions at 0.1 M stabilizes DNA by 3.2 kcal/mole P at 25 °C. At 10^{-3} M the corresponding stabilization is twice as great.

C. Colligative properties

Since, as described in II, the free energy of a polyelectrolyte solution is theoretically calculable, expressions may also be obtained for all the observables associated with classical equilibrium electrochemistry such as activity coefficients, osmotic coefficients, Donnan effects and heats of solution and mixing (Manning, 1972*a*, 1974; Devore & Manning, 1974; Boyd, Wilson & Manning, 1976; Manning, 1976*a*). As an example let

TABLE 3. *Correlation of activity coefficient data for MgDNA/MgCl₂/H₂O*

<i>X</i>	<i>c_P</i>	$(\gamma_{\text{Mg}}/\gamma_{\text{Mg}}^0)_{\text{obs}}^\dagger$	$(\gamma_{\text{Mg}}/\gamma_{\text{Mg}}^0)_{\text{calc}}^\ddagger$
1.2	1.0×10^{-2}	0.47	0.50
1.4	6.1×10^{-4}	0.36	0.46
2.5	2.6×10^{-3}	0.35	0.34
3.1	1.0×10^{-2}	0.29	0.30
3.1	6.2×10^{-4}	0.25	0.30
6.1	2.6×10^{-3}	0.21	0.21
6.2	1.0×10^{-2}	0.20	0.20
7.7	6.4×10^{-4}	0.16	0.18
12.2	2.6×10^{-3}	0.14	0.15
15.4	6.5×10^{-4}	0.13	0.13
∞	5×10^{-4} to 1×10^{-2}	0.075 ± 0.05	0.072

† Data from Lyons & Kotin (1965).

‡ Calculated from equation (19) of text.

us consider the activity coefficient γ_{Mg} of the Mg^{2+} ion in the ternary solution $\text{MgDNA}/\text{MgCl}_2/\text{H}_2\text{O}$. Theoretical activity coefficients are easily obtained, since the number of associated counterions, which do not contribute to the apparent activity, is known, while unassociated counterions contribute to simple Debye-Hückel screening interactions. The theoretical formula for γ_{Mg} is (Kwak, O'Brien & MacLean, 1975),

$$\ln(\gamma_{\text{Mg}}/\gamma_{\text{Mg}}^0) = -\frac{1}{4}\xi^{-1}X(\frac{1}{2}\xi^{-1}X + \frac{3}{2})^{-1} + \ln[(\frac{1}{2}\xi^{-1}X + 1)(X + 1)^{-1}]. \quad (19)$$

In this formula γ_{Mg}^0 is the activity coefficient of Mg^{2+} in the binary solution $\text{MgCl}_2/\text{H}_2\text{O}$ at the same MgCl_2 concentration as in the polyelectrolyte solution, X is the ratio of polyelectrolyte equivalent concentration (normality) to MgCl_2 equivalent concentration, and ξ is the charge density parameter defined by equation (1) and equal to 4.2 for DNA in an aqueous environment.

Table 3 presents a comparison of observed values of γ_{Mg} with values calculated from equation (19). X has been defined above, c_P is the DNA normality (molarity of phosphate groups) and MgCl_2 normalities are equal to $c_P X^{-1}$. The last entry in the table, $X = \infty$, refers to a pure MgDNA solution in the absence of added MgCl_2 . Note the remarkably good agreement between observed and theoretical values of $\gamma_{\text{Mg}}/\gamma_{\text{Mg}}^0$ over the entire composition range, $1 < X < \infty$ and MgCl_2 normalities up to 0.01 N. Underlying the predicted values is the concentration-

invariant theoretical charge fraction $(2\xi)^{-1} = 0.12$; Table 3 adds to the evidence for its physical reality.

D. Transport properties

It is an easy matter to calculate the electrophoretic mobility U expected for the polyelectrolyte model under discussion, namely a linear array of effective point charges of magnitude $N^{-1}\xi^{-1}q$ and spacing b . The starting point is Kirkwood's theory of the translational friction coefficient for macromolecules composed of identical subunits (Bloomfield, Crothers & Tinoco, 1974). It need only be recognized that the hydrodynamic interaction between each pair of subunits (in this case, point charges) is attenuated, not only by the familiar inverse dependence on distance but also by the oppositely charged Debye-Hückel atmosphere that surrounds each fixed charge and moves in the opposite direction, dragging the solvent with it, under the action of an applied electric field. The quantitative expression of this added attenuation is the appearance of a Debye screening factor $\exp(-\kappa b|i-j|)$ in the hydrodynamic interaction between fixed charges i and j . The summation over all pairs is then exactly the same operation as summing the Debye-Hückel pair potentials of equation (3), and the final result is

$$U = N^{-1}(\epsilon kT/900\pi\eta q) |\ln \kappa b|, \quad (20)$$

where η is the viscosity and all other quantities have been previously defined. At 0 °C in water the numerical value of the factor in parentheses is 1.37×10^{-4} ; the units of U are standard, $\text{cm}^2 \text{sec}^{-1} \text{V}^{-1}$.

Mobilities predicted from equation (20) are compared in Table 4 with values measured for DNA ($b = 1.7 \times 10^{-8} \text{ cm}$) in excess MgCl_2 ($N = 2$) at 0 °C. It should be noted that, although prediction consistently overestimates observation, the ratio of calculated to observed values is almost constant over a range of MgCl_2 normalities n_{MgCl_2} that cover more than two orders of magnitude. The data are therefore in reasonable agreement with the predicted linear dependence on the logarithm of ionic strength, and the discrepancy in absolute values can probably be ascribed to the obviously oversimplified description of the hydrodynamic particle as a linear array of points. Nevertheless, the *electrostatic* contribution to the mobility is apparently, as anticipated, accurately contained in this model. The theoretical mobility explicitly rests on a concentration-invariant charge fraction $(N\xi)^{-1}$, which therefore is supported by the mobility data in Table 4.

TABLE 4. *Comparison of predicted and observed mobilities*

n_{MgCl_2}	κ^{-1} (Å)	$10^4 U_{\text{obs}}^\dagger$	$10^4 U_{\text{calc}}^\dagger$	$U_{\text{calc}}/U_{\text{obs}}'$
5.6×10^{-4}	106.2	1.6 ± 0.2	2.83	1.77
1.0×10^{-3}	79.5	1.5 ± 0.2	2.63	1.75
5.2×10^{-3}	34.9	1.00	2.07	2.07
1.1×10^{-2}	24.0	0.86	1.81	2.10
3.2×10^{-2}	14.1	0.76	1.45	1.91
6.2×10^{-2}	10.1	0.60	1.22	2.03
1.9×10^{-1}	5.8	0.45	0.84	1.87

† Data from Papon & Strauss (unpublished).

‡ Calculated from equation (20) of text.

E. The dissociation-field effect

The extent of counterion condensation, $N^{-1}(1 - N^{-1}\xi^{-1})$ moles condensed counterion of valence N per polyelectrolyte equivalent, is invariant to changes of ionic strength. Application of a strong electric field E , however, can cause partial dissociation of the condensed counterions, thereby increasing the charge fraction of the polyelectrolyte. The dissociation-field effect is well known in the area of partially associated simple electrolyte solutions, since the linear increase with E of the conductance of such solutions is accepted as a direct measure of an increased dissociation constant (Onsager, 1934; Onsager & Liu, 1965). The conductance of polyelectrolyte solutions also increases linearly with E , but it is not clear at present that the field-dissociation effect is the dominant factor in this behaviour. It is conceivable that charge fractions can be directly measured in the presence of a field, as they have been in its absence, but no attempt in this direction has yet been undertaken. Therefore the field-dissociation effect for polyelectrolytes cannot yet be said to have been directly observed. The great current interest in the effect is due rather to observations that sufficiently strong electric fields can induce conformational changes in ordered polynucleotide structures (Pörschke, 1976*b*; Revzin & Neumann, 1974).

The theory for polyelectrolytes (Manning, 1977*c*) is surprisingly easy, in view of the almost legendary difficulty of Onsager's 1934 paper on small electrolytes. At zero field the contribution to the free energy from condensed counterions is, aside from the effective decreased charge on the polyion, an ideal mixing entropy. When the field E (in e.s.u.) is turned on, the free energy is supplemented by a work term proportional

to $NqaE$, where Nq is the charge of a counterion and a is a distance. That is, in order to remain condensed, a counterion must do work against the field which tends to tear it from the polyion. The key to the derivation is the unambiguous knowledge of a , which is simply the cylindrical radius of the free volume V_P determined by equation (13). The average position of a counterion, if condensed, is on the cylindrical axis of the free volume; to remain there, the counterion must work against the field over the distance a , past which it is no longer condensed. Values of a have been tabulated as a function of ξ (Manning, 1977*a*); for DNA, $\xi = 4.2$ and $a = 14.2$ Å for univalent counterions.

With this added work term condensed counterions have higher energy; hence at the free energy minimum there will be fewer condensed counterions than for the field-free case. Define a degree of dissociation α by

$$\theta_N(E)/\theta_N(0) = 1 - \alpha, \quad (21)$$

where $\theta_N(E)$ is the fraction of a condensed counterion per polyelectrolyte charge at field E ; $\theta_N(0)$, the corresponding field-free quantity, has already been found and is given by equation (12). The relation between α and E at the free energy minimum is

$$\bar{E} = \alpha[1 - 2(N\xi - 1) \ln \kappa b], \quad (22)$$

where

$$\bar{E} = (\frac{1}{2}\pi) (NqaE/kT). \quad (23)$$

For an aqueous polyelectrolyte solution at 25 °C containing excess 1 : 1 salt of ionic strength 10^{-2} M, a very high applied field of 402 kV/cm produces only 10% dissociation ($\alpha = 0.1$) if $\xi = 4$ (the approximate value for DNA). At the higher ionic strength 10^{-1} M, dissociation requires less work because the dissociated ions enter a more effective screening atmosphere; the field corresponding to $\alpha = 0.1$, however, is still 262 kV/cm for $\xi = 4$. Although the extent of counterion condensation is the *same* at 0.01 and 0.1 M in the absence of a field, it may be said that the condensed counterions are held more loosely at the higher ionic strength, in the sense that the field required to tear some of them away is then not as high as in the more dilute solution. In Section VD the theory is applied to the effect of a strong field on the melting of DNA.

F. Specific ionic effects

To a slight extent, DNA in aqueous solution has been observed to

discriminate among cations of the alkali metal series, preferentially selecting Li^+ over Na^+ and K^+ in differential dialysis experiments (Span & Strauss, unpublished) and exhibiting higher T_m 's in Li^+ solution relative to Na^+ and K^+ (Gruenwedel, Hsu & Lu, 1971). Since these ions are unable to discriminate between A·T-rich and G·C-rich DNA (Shapiro *et al.* 1969*b*) there is no reason to suspect groove binding, and the almost unrestricted movement of bound Na^+ and Rb^+ (Reuben *et al.* 1975) suggests that varying degrees of 'tightness' of localized binding to phosphate groups is not an appropriate explanation. We have seen that the local concentration of condensed univalent cations in the immediate neighbourhood of DNA is about 1.2 M, regardless of the bulk ionic strength. Moreover, Li^+ is more highly hydrated (i.e. binds *water* more tightly) than Na^+ or K^+ (Robinson & Stokes, 1959). Given the sensitivity of DNA structure to solvent effects, it would be surprising if DNA in a local environment of 1 M aqueous M^+ did *not* exhibit at least small differential effects depending on the species M^+ . By the same token circular dichroism variations upon addition of Mg^{2+} (Clement *et al.* 1973) are perhaps also not an unexpected consequence of changes in local solvent properties, since Mg^{2+} binds *water* extremely tightly (Eigen, 1963).

For one such effect a quantitative estimate can be made. If it is assumed that the electrophoretic mobility of DNA is inversely proportional to the *local* solvent viscosity (Abramson, Moyer & Gorin, 1942), then one would predict that mobilities at any ionic strength in aqueous MCl should vary in inverse proportion to the viscosities of 1 M-MCl (the Cl^- ion contributes little to the ionic effect on viscosity). On this basis DNA mobilities would be expected to vary in the proportion 1.23 : 1.13 : 1.00 in aqueous K^+ , Na^+ and Li^+ , respectively, at 0 °C. The observed variation (Ross & Scruggs, 1964) is 1.37 : 1.16 : 1.00.

A recent development of possibly great importance is the discovery that half of the Na^+ counterions are coordinated to $\text{O}_{(2)}$ of uracil and thymine in *crystalline* ApU and pTpT, respectively (Seeman *et al.* 1976; Camerman, Fawcett & Camerman, 1976). The $\text{O}_{(2)}$ atom of thymine and uracil is a carbonyl oxygen exposed in the minor groove of double helical structures and not involved in base pair hydrogen binding. Although all current evidence *in solution* points to the absence of groove binding for Na^+ (presumably because its electric field intensity is more easily compensated by complete hydration in solution than in even a relatively highly hydrated crystal), the $\text{O}_{(2)}$ atom could be an

important factor in the apparent ability of the A · T base pair to bind more water in solution than the G · C pair (Tunis & Hearst, 1968) as well as in the small selective affinity of the tetramethyl- and tetraethylammonium cations for A · T-rich regions of DNA relative to G · C-rich regions (Shapiro *et al.* 1969; Melchior & von Hippel, 1973).

Another interesting result is the charge reversal of DNA in aqueous MgCl_2 if the concentration is in excess of 1 N- MgCl_2 (Papon & Strauss, unpublished). DNA then behaves like a cation in an electrophoresis experiment. It would be of great interest to know if a conformational change, possibly to a compact form, is associated with this phenomenon. The constant anionic charge fraction 0.12 mentioned repeatedly in previous sections, of course, refers to an apparently much simpler situation at lower Mg^{2+} concentrations.

IV. THEORY OF DELOCALIZED BINDING OF N -VALENT CATIONIC SUBSTANCES TO POLYNUCLEOTIDES IN AQUEOUS EXCESS 1 : 1 SALT

A. Introduction

The subject of this section is a statistical mechanical analysis of binding constants obtained under typical experimental conditions. The goal is the *a priori* prediction of binding constants, properly defined, under the assumption that the bound cationic substance is delocalized (free translation near the polynucleotide surface). Comparison with measured values will then provide suggestive information about the molecular structure of the cation/polynucleotide complex. The central role of the counterion diffusion potential will be stressed, and the reader may wish to refer to the last paragraph of Section II C before proceeding.

B. One-variable theory

1. *Minimization of the free energy.* I shall begin with a discussion of binding relations obtained by treating the free energy as a function of the single variable θ_N , the number of M^{N+} cations bound per phosphate group. As in Section II C, the excess salt condition, equation (7), is assumed. In addition the inequality

$$c_1 \gg c_N, \quad (24)$$

where c_N is the molarity of *free* N -valent cations and c_1 the molarity of supporting 1 : 1 salt, proves useful without imposing a severe restriction

on applications to commonly employed binding conditions. It allows us to retain equation (5) for the Debye screening constant κ , since c_1 is then effectively equal to the ionic strength I . It also motivates the modelling specification that V_P as given by equation (15) for 1:1 salts, defines the region of binding for both univalent and N -valent cations. The numerical values of V_P for several polynucleotides may be found in Table 1.

The one-variable approach turns out to require two separate cases, low binding levels (small θ_N) and high levels (θ_N approaching saturation). The two-variable theory developed subsequently enables one to avoid this separation and also to assess the approximations inherent in one-variable minimization. On the other hand the one-variable theory will provide a simple and attractive interpretation for the otherwise physically obscure two-variable results.

For low θ_N Table 1 motivates the assumption that θ_1 , the number of bound M^+ cations per phosphate, is fixed at the value

$$\theta_1 = 1 - \xi^{-1} - N\theta_N, \quad (25)$$

that is, the binding of one M^{N+} ion results in the release of N bound M^+ ions, so that the charge fraction of the polynucleotide (fraction of phosphate charge uncompensated by bound cations) remains invariant at $1 - \theta_1 - N\theta_N = \xi^{-1}$. It is this restriction on θ_1 that allows reduction of the problem to the minimization of the free energy with respect to the single variable θ_N . From equation (25) it follows that the calculation for low θ_N is mathematically applicable to the range

$$0 \leq \theta_N \leq N^{-1}(1 - \xi^{-1}). \quad (26)$$

With the notation of equation (6) the electrostatic free energy required to charge the phosphate groups is

$$\bar{g}_{el} = -\xi^{-1} \ln(1 - e^{-\kappa b}), \quad (27)$$

obtained by replacing the charge fraction $1 - N\theta_N$ in equation (8) by ξ^{-1} , in accordance with the discussion of equation (25). This contribution to the free energy is independent of θ_N ; hence the electrostatic repulsion does not determine, even partially, the extent of binding of the M^{N+} ions, at least in the binding range (26) and subject to the constraint equation (25).

The only other contributions to the free energy of the system are

assumed to be from the ideal mixing of the various species,

$$\begin{aligned}\bar{g}_{\text{mix}} = & N\theta_N \ln(10^{-3}c_1) + \theta_1 \ln(\theta_1/V_P) \\ & + (c_N/c_P) \ln(10^{-3}c_N) + \theta_N \ln(\theta_N/V_P) + \theta_N,\end{aligned}\quad (28)$$

where terms independent of θ_N have not been included. The first term on the right-hand side of equation (28) is the free energy of mixing of the free univalent cations M^+ , with recognition that $N\theta_N M^+$ ions per phosphate have been freed from the bound state if the extent of binding of the N -valent molecule is θ_N . The second term is the mixing contribution of the $\theta_1 M^+$ ions remaining bound; θ_1 is given by equation (25) as a function of θ_N . The third term is contributed by free M^{N+} ions of concentration c_N , which satisfies the condition $dc_N/d\theta_N = -c_P$ if the *total* amount of M^{N+} ions is specified as fixed (corresponding to the typical experiment of adding a given amount of M^{N+} to the system, measuring the amount bound or free, and obtaining the amount free or bound by difference). The fourth term is the free energy of mixing of bound M^{N+} ions, assumed delocalized just like the univalent ions. The fifth term combines the mixing entropy of solvent with a higher-order term from the mixing of free M^+ . Further detail for the case $N = 2$ may be found in Manning (1977*b*). Minimization of $\bar{g}_{\text{mix}}$ as a function of θ_N yields the equilibrium binding relation

$$\begin{aligned}\log K_N = & \log(V_P/1000e) \\ & + N \log[(1000e/V_P)(1 - \xi^{-1} - N\theta_N)c_1^{-1}],\end{aligned}\quad (29)$$

where

$$K_N = \theta_N/c_N. \quad (30)$$

The quantity K_N is proportional to the equilibrium constant for the transfer of an M^{N+} ion from free solution to the 'bound' region V_P , and, as such, serves as the 'association constant' for the binding of M^{N+} . Recall also that K_N as defined by equation (30) is the ordinate in the usual Scatchard plot and is a directly accessible experimental quantity. The ionic strength dependence of K_N is given by the coefficient

$$d \log K_N / d \log c_1 = -N, \quad (31)$$

while, as a function of the extent of prior binding θ_N , K_N is a monotone decreasing N th order polynomial with a single zero at $\theta_N = N^{-1}(1 - \xi^{-1})$.

In Fig. 1 K_2 is represented by a Scatchard plot; the dashed curve for $\theta_2 \leq 0.38$, a number corresponding to $\xi_{\text{DNA}} = 4.2$, is equation (29) in the range specified by (26). The additional information in Fig. 1 and the other figures to which reference is made at this point pertain to calcula-

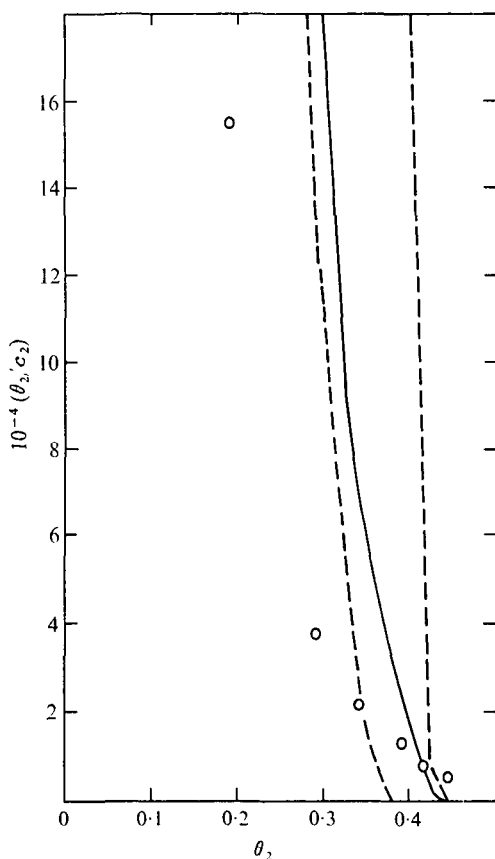


Fig. 1. Scatchard representation of divalent ion binding. Dashed curve for low θ_2 , equation (29); dashed curve for high θ_2 , equation (36); solid curve, equations (53) and (54). Data points from Clement *et al.* (1973). Ionic strength, 0.001 M.

tions and data that are discussed subsequently. In each of these figures only the dashed curves for $\theta_2 \leq 0.38$ are germane to the present case. Fig. 2 gives a logarithmic representation of K_2 . Fig. 3 shows the ionic strength dependence of K_2 , equation (31). Fig. 4 shows how the bound fraction of M^+ cations decreases as M^{2+} is progressively bound, in this case following equation (25). Finally, Fig. 5 gives the total fraction of the phosphate charge compensated by bound cations of both species M^+ and M^{2+} as a function of θ_2 ; for the present case, as mentioned in connection with equation (25), this number is given by $1 - \xi^{-1}$ (equal to 0.76 for DNA) for $\theta_2 \leq \frac{1}{2}(1 - \xi^{-1})$.

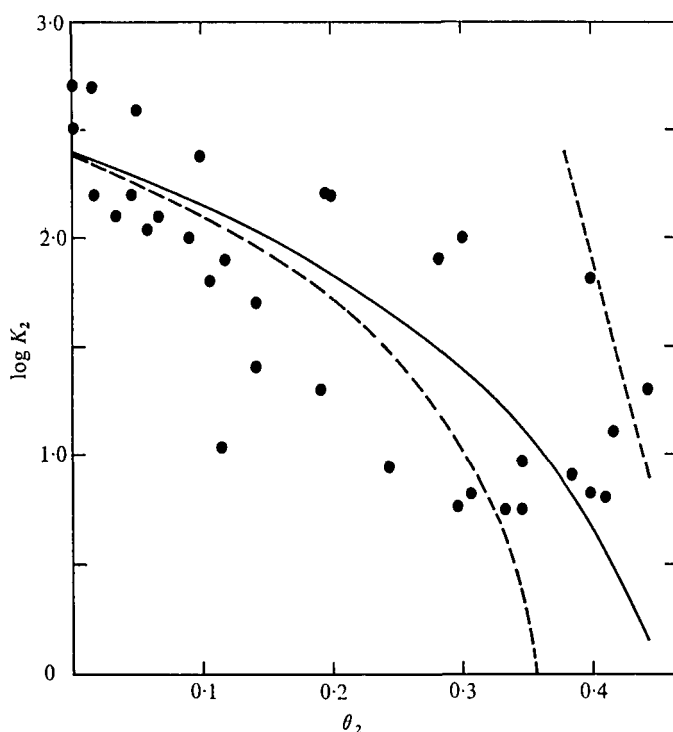


Fig. 2. Logarithmic representation of divalent ion binding. Curves as for Fig. 1; data from Archer *et al.* (1972), Clement *et al.* (1973), Bhat (1974), Skerjanc & Strauss (1968), and Zubay & Doty (1958). Curves are drawn for 0.1 M ionic strength; points represent ionic strengths in the range 0.001–0.2 M but are normalized to 0.1 M according to procedure described in text.

For the binding region

$$\theta_N \geq N^{-1}(1 - \xi^{-1}) \quad (32)$$

a separate calculation is required. In this region the constraint that reduces the problem to one variable is

$$\theta_1 = 0, \quad (33)$$

that is, all bound M^+ cations have been displaced by M^{N+} . Note that equation (33) is continuous with equation (25) at the mutually included point $\theta_N = N^{-1}(1 - \xi^{-1})$. Now

$$\bar{g}_{el} = -(1 - N\theta_N)^2 \xi \ln(1 - e^{-\kappa b}), \quad (34)$$

in analogy to equation (8), while

$$\bar{g}_{mix} = (c_N/c_P) \ln(10^{-3}c_N) + \theta_N \ln(\theta_N/V_P) + \theta_N, \quad (35)$$

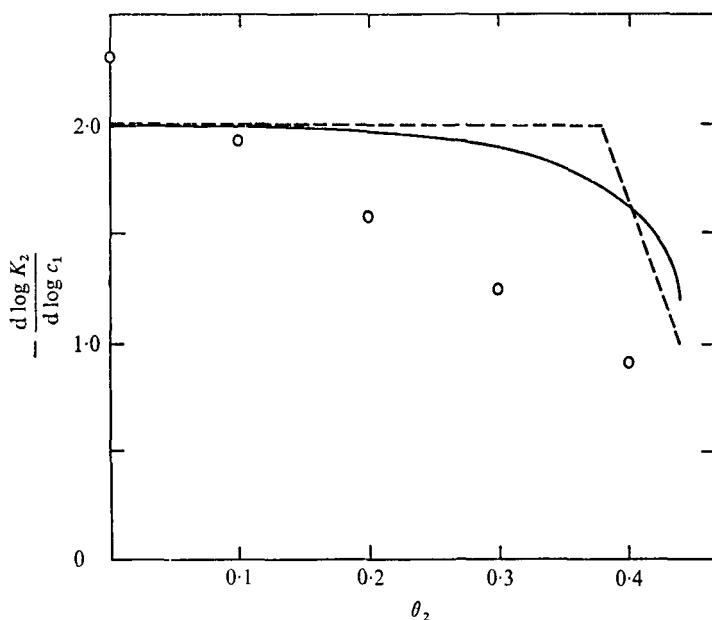


Fig. 3. Ionic strength dependence of divalent ion binding. Dashed curve for low θ_2 , equation (31); dashed curve for high θ_2 (after break), equation (37); solid curve computed from equations (53) and (54). Data from Krakauer (1974) based on measurements in the range 0.01–0.1 M.

since only the mixing of free M^{N+} , bound M^{N+} , and solvent depend on θ_N . Minimization of $\bar{g}_{el} + \bar{g}_{mix}$ with respect to θ_N , with recognition that $dc_N/d\theta_N = -c_P$, produces the following relation for the value of θ_N at equilibrium:

$$\log K_N = \log (V_P/1000) - 2N\xi(1 - N\theta_N) \log \kappa b, \quad (36)$$

where K_N is defined by equation (30) and κ is conveniently given by equation (5). The approximation $1 - e^{-\kappa b} \approx \kappa b$ has been incorporated into (36) because (35), with which (33) is continuous, is motivated by the behaviour at low concentration illustrated in Tables 1 and 2. For the ionic strength dependence of K_N , it follows from (5) and (36) that

$$d \log K_N / d \log c_1 = -N\xi(1 - N\theta_N), \quad (37)$$

a relation that preserves continuity with (31) at $\theta_N = N^{-1}(1 - \xi^{-1})$. As a function of θ_N , K_N decreases exponentially like $\exp [2N^2\xi \ln (\kappa b) \theta_N]$.

These relations are illustrated for K_2 and DNA by the dashed curves in Figs. 1–5 corresponding to $\theta_2 \geq 0.38$. In Fig. 4, of course, the dashed

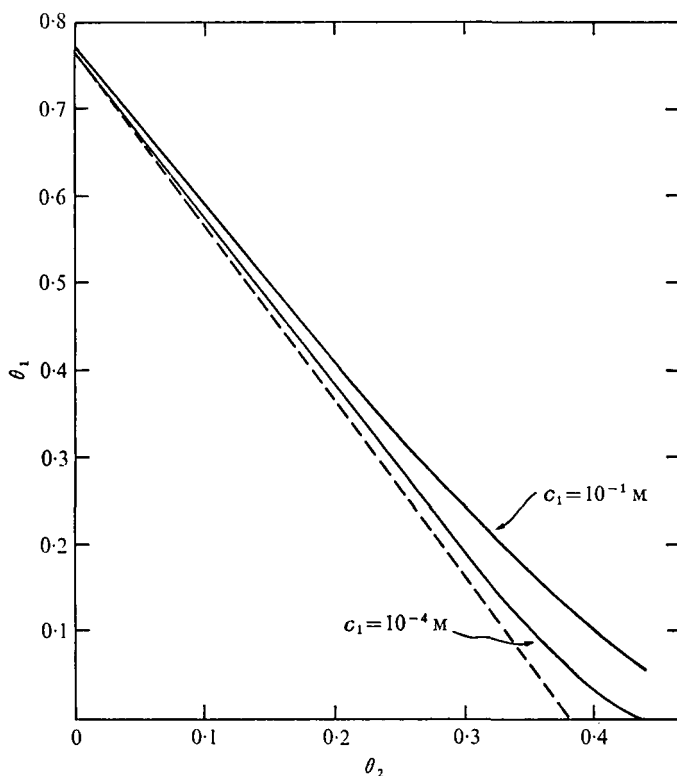


Fig. 4. Fractional number (per phosphate) of bound univalent counterions as a function of fractional number of bound divalent ions. Dashed curve, equation (25). Solid curves computed from equations (53) and (54) for the indicated ionic strengths.

curve in this range coincides with the abscissa, in accordance with (33). An important feature of the figures not yet mentioned is the existence of a maximum value $\theta_N^{\max}$. Due to the constraint imposed by inequality (24), $\theta_N^{\max}$ cannot be obtained from (36), which yields a positive value of K_N for any value of θ_N . However, as M^{N+} continues to be added to the system, so that inequality (24) becomes invalid and then reversed, the case that eventually presents itself is that of the polynucleotide in an aqueous environment of M^{N+} cations with negligible amounts of M^+ . This situation is the same as that discussed in Sections II and III. According to (17), then,

$$\theta_N^{\max} = N^{-1}(1 - N^{-1}\xi^{-1}) \quad (38)$$

and the binding range to which (36) is restricted is defined by inequality

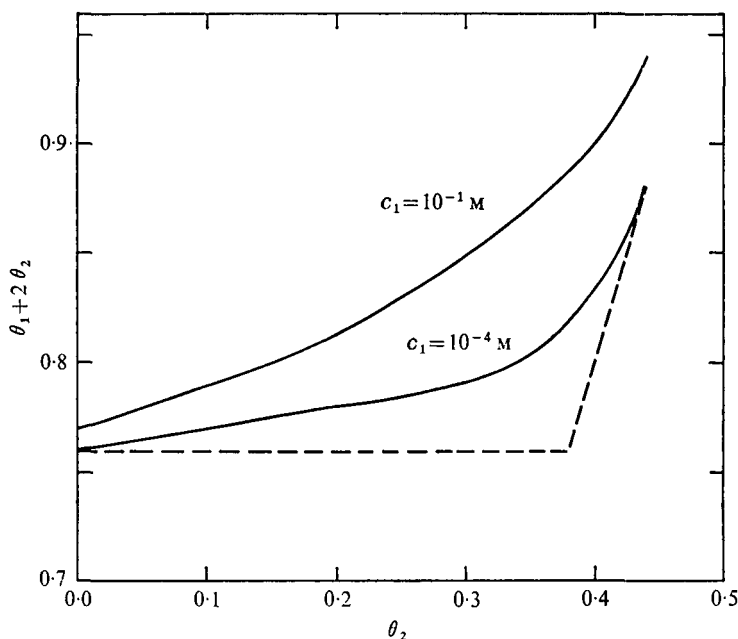


Fig. 5. Fraction of phosphate charge compensated by the combined charge of bound univalent and divalent ions, i.e. $\theta_1 + 2\theta_2$. Dashed curve for low θ_2 from equation (25); dashed curve for high θ_2 from equation (33) (i.e. this curve is merely $2\theta_2$ v. θ_2); solid curves computed from equations (53) and (54) for the indicated ionic strengths.

(32) and (38). For DNA, $\theta_2^{\max} = 0.44$, a value that has been closely confirmed by measurement for Mg^{2+} , as discussed in Sections II and III. Recall from IIIF that this value is not an absolute upper limit; rather, it may represent a saturation plateau that is eventually exceeded, but not before concentrations of about 1 M free Mg^{2+} have been attained and factors other than electrostatic may become dominant.

2. *Chemical model.* Additional insights may be obtained by formulating the binding process as a chemical reaction,



and applying to it the standard thermodynamic analysis of chemical reactions. The reaction affinity (Kirkwood & Oppenheim, 1961) $\tilde{\Delta}\mu_N$ for (39) is

$$\tilde{\Delta}\mu_N = \mu_{N,b} - \mu_{N,f}, \quad (40)$$

where $\mu_{N,b}$ and $\mu_{N,f}$ are the chemical potentials of bound and free M^{N+} , respectively. By appropriate differentiation of the free energy of

the system for low binding levels, (28), the required chemical potentials, in units of RT , may be obtained as

$$\mu_{N,f} = \ln 10^{-3} c_N, \quad (41)$$

$$\mu_{N,b} = -(N-1) + N \ln [10^{-3} c_1 V_P (1 - \xi^{-1} - N\theta_N)^{-1}] + \ln (\theta_N/V_P). \quad (42)$$

Note that the local and free concentrations of univalent M^+ are 'hidden variables' relative to reaction (39) and appear in the 'standard state' component of $\mu_{N,b}$ (first two terms in r.h.s. of (42)). From (40), (41), (42), (16) and (25) one has

$$\tilde{\Delta}\mu_N = \tilde{\Delta}\mu_N^0 + \ln (c_N^{00}/c_N), \quad (43)$$

where

$$\tilde{\Delta}\mu_N^0 = -(N-1) + N \ln (c_1/c_1^{100}). \quad (44)$$

In (43) the local molarity c_N^{100} of bound M^{N+} is given in analogy to (16),

$$c_N^{100} = 10^3 \theta_N / V_P. \quad (45)$$

It is easily verified that the equilibrium condition $\tilde{\Delta}\mu_N = 0$ yields the equilibrium binding relation, (29).

The equilibrium constant for reaction (39) is conventionally defined by

$$\bar{K}_N = (c_N^{100}/c_N)_{\text{equil}} = \exp (-\tilde{\Delta}\mu^0), \quad (46)$$

where the 'bar' on $\bar{K}_N$ distinguishes it from the quantity K_N of (30), to which it is proportional. Then, with (44),

$$\bar{K}_N = e^{N-1} (c_1^{100}/c_1)^N. \quad (47)$$

The association constant $\bar{K}_N$ for binding of M^{N+} is large if the concentration c_1 of free M^+ (ionic strength) is appreciably less than the local concentration c_1^{100} of bound M^+ . This conclusion is in complete agreement with the concept of the 'potentiated' polyelectrolyte discussed in IIC; reaction (39) is driven by the diffusional entropy of the 'hidden' process $NM^+(c_1^{100}) \rightarrow NM^+(c_1)$. The constant contribution $(N-1)$ to $\ln \bar{K}_N$ arises from the osmotic inactivity of the bound cations, which, although free to move on the surface of the polyion, travel with the polyion and hence do not independently lower the activity of the solvent.

Note that, when $N = 1$, the equilibrium condition $\tilde{\Delta}\mu = 0$ reduces to

$$c_N^{100}/c_N = c_1^{100}/c_1 \quad (N = 1), \quad (48)$$

a relation that underscores the complete absence of specific effects in the model; selectivity is on the basis of valence alone.

The chemical thermodynamical model allows a systematic calculation of the free energy ΔG stabilizing the M^{N+} /polynucleotide complex,

$$\Delta G/RT = \int_0^{\lambda_{eq}} \tilde{\Delta\mu}_N d\lambda. \quad (49)$$

In this equation ΔG is the free energy of the system at equilibrium relative to its free energy at the moment before commencement of reaction (39), when all M^{N+} ions are present and mixed uniformly but none are yet bound. The quantity λ is the progress variable, or degree of advancement of the reaction, and λ_{eq} is its value after the reaction has proceeded to equilibrium (Kirkwood & Oppenheim, 1961). For reaction (39), λ is the number of moles of bound M^{N+}

$$\lambda = \theta_N n_P, \quad (50)$$

where n_P is the number of moles of phosphate groups in the system. After the integration in (49) is carried through, extensive cancellation may be achieved by application of the relation $\tilde{\Delta\mu}(\lambda_{eq}) = 0$ (equivalent to equation (29)), and the final result is

$$\Delta\bar{g} = \theta_N^t \ln [1 - (\theta_N/\theta_N^t)] + N\theta_N + (1 - \xi^{-1}) \ln [(1 - \xi^{-1} - N\theta_N)/(1 - \xi^{-1})], \quad (51)$$

where it is understood that θ_N is the equilibrium value determined by (29) and (30), θ_N^t is the total number of moles of M^{N+} present in the system per mole phosphate, and $\bar{g}$ was defined by (6).

To illustrate this relation, suppose that the phosphate (base) molarity is 10^{-4} M at an ionic strength $c_1 = 0.1$ M. For the equilibrium extent of binding $\theta_2 = 0.25$ (50% phosphate charge compensation), (29) yields $\theta_2^t = 87$ or a total divalent cation molarity 8.7×10^{-3} M. One then calculates the value $\Delta\bar{g} = -0.56$ from (51) for the B DNA value $\xi = 4.2$. At 25 °C, then, the loss of 0.33 kcal free energy/mole phosphate accompanies the binding of 0.25 mole divalent cation/mole phosphate under these conditions. An interesting comparison is that, at the same ionic strength in the absence of divalent metal ion, formation of the bound layer of univalent cations is stabilized by the much larger amount 3.2 kcal/mole phosphate, as discussed in III B.

C. Two-variable theory†

The constraints on θ_1 , equation (25) for low binding levels and (33) for high, may be relaxed in a straightforward manner. The electrostatic free energy is taken as

$$\bar{g}_{el} = -(1 - \theta_1 - N\theta_N)^2 \xi \ln(1 - e^{-\kappa b}), \quad (52)$$

that is, both θ_1 and θ_N are variable and the net charge density after binding of univalent and N -valent cations is reduced from the stoichiometric value based on the phosphates alone by the factor $(1 - \theta_1 - N\theta_N)$. The mixing free energy is identical to equation (28), except that the coefficient $N\theta_N$ in the first term of the r.h.s. of (28) must be replaced by $-\theta_1$ to account for variable θ_1 . Minimization of $\bar{g}_{el} + \bar{g}_{mix}$ with respect both to θ_1 and θ_N leads to the following two equations for the equilibrium values of θ_1 and θ_N :

$$1 + \ln(10^3 \theta_1 V_P^{-1}/c_1) = -2\xi(1 - \theta_1 - N\theta_N) \ln(1 - e^{-\kappa b}), \quad (53)$$

$$\ln(\theta_N/c_N) = \ln(V_P/1000e) + N \ln[(1000e/V_P) c_1^{-1} \theta_1]. \quad (54)$$

It is instructive to use these equations first to regain the results of the one-variable theory. Following the analysis of (11), one sees that (25) is indeed an approximate solution of (53) for small θ_N . Substitution of (25) into (54) yields a binding relation for M^{N+} identical to (29). Thus, the one-variable result for low levels of binding, (25) and (29), is an approximate solution of (53) and (54) for small θ_N .

For values of θ_N near saturation, an approximate solution of (53) and (54) is obtained as follows. First, note that

$$V_P = 1000e(1 - \xi^{-1})(\kappa b)^2 c_1^{-1} \quad (55)$$

as a consequence of (11) with $N = 1$ and (14). Equation (55) is simply another form of (15). Now, anticipating that θ_1 is small near saturation of M^{N+} binding, substitute $\theta_1 = 0$ into the r.h.s. of (53), along with the approximation $1 - e^{-\kappa b} \approx \kappa b$. With (55) the result is an improved approximation for θ_1 ,

$$\theta_1 \approx (1 - \xi^{-1})(\kappa b)^{2[1 - \xi(1 - N\theta_N)]}. \quad (56)$$

Substitution of this expression for θ_1 into (54), followed by a second application of (55), yields a result identical to (36), the prediction of one-variable theory for high levels of binding. Thus, (33) and (36)

† This section represents work done in collaboration with Nancy de Marky.

constitute an approximate solution of equations (53) and (54) for high binding levels of M^{N+} .

Equations (53) and (54) can be solved exactly by numerical iteration. It is convenient to fix θ_N , obtain θ_1 from equation (53) by iteration, and then substitute θ_1 into (54) to get $K_N = \theta_N/c_N$ corresponding to the fixed value for θ_N . (An n th iterate for θ_1 may be obtained by substituting the $(n-1)$ st into either the left- or right-hand side of (53); the scheme that converges depends on the value of θ_N . For small θ_N , a convenient zeroth-order trial for θ_1 is obtained from (25); for large θ_N , an appropriate starting point is $\theta_1 = 0$.)

The numerical results for DNA ($\xi = 4.2$, $V_P = 646 \text{ cm}^3$; see Table 1) are presented as the full curves in Figs. 1–5, the meanings of which were discussed above in connexion with the one-variable theory (dashed lines). At this point it can be remarked that the one-variable theory has provided useful approximations. Equation (29) is fairly accurate over a broad part of the low-binding end of the range (and nearly exact at $\theta_N = 0$), and (36) is reasonably good right at saturation (it is appreciably better at saturation than indicated by Fig. 2 if ionic strengths lower than 0.1 M are considered). More important than the degree of numerical accuracy of the approximations, since (53) and (54) are easily solved exactly, is the physical insight provided by the one-variable theory and discussed in detail above. The exact solution of (53) and (54) validates that discussion and sets limits to its applicability.

D. Divalent metal ion binding

1. *Summary of results for $N = 2$.* A summary of the main theoretical results obtained here must begin with the definition of the association constant K_2 (see equation (30)) as proportional to the equilibrium constant for the transfer of a divalent metal ion from free solution to a local volume region surrounding the polynucleotide, within which the metal cations move freely. For a duplex helix the ‘association region’ or ‘free volume’ V_P extends about $7 \text{ \AA}$ out from the surface of the helix (see Table 1). Thus defined, $\log K_2$ is a linear function of the logarithm of the ionic strength (i.e. concentration of univalent cations from the excess 1 : 1 supporting electrolyte) with slope nearly equal to -2 over most of the binding range but dropping off rapidly to -1 as saturation is approached.

The association is anticooperative; K_2 decreases as the binding level increases. Over most of the binding range the functional dependence of

K_2 on extent of prior binding θ_2 is intrinsically quadratic (see equation (29)), becoming exponential only at the extremity near saturation. The origin of the anticooperativity is entropic; c_1^{lo} is decreased when divalent cation binds, in accordance with (25), so that the driving force decreases with θ_2 ; see (47). The repulsive interactions among bound divalent metal ions usually invoked to account for anticooperative behaviour are assumed here to contribute negligibly to the free energy and are not present in the model.

Equation (47), valid over most of the binding range, states that nearly the entire thermodynamic force that drives divalent cation binding is entropic, consisting of the positive concentration difference between bound but delocalized univalent counterions and univalent counterions free in the solution bulk. Binding of a divalent ion is accompanied by the net flow of two bound univalent counterions along this gradient, out into free solution. Only near the saturation end of the binding range does the decrease of electrostatic free energy (predominately still an entropy term, incidentally, since ϵT is nearly independent of temperature for water) become a source of binding free energy.

Substitution of the numerical values $\xi_{DNA} = 4.2$ and $V_P^{DNA} = 646 \text{ cm}^3$ from Table 1 into equation (29) gives a good numerical approximation for K_2^{DNA} for θ_2 less than about 0.30,

$$\log K_2^{DNA} = 0.39 + 2 \log (1 - 2.63\theta_2) - 2 \log c_1. \quad (57)$$

The exact equations (53) and (54) may be solved numerically, however, over the entire binding range.

2. *Comparison of theory with Mg^{2+} binding data.* Table 5 presents a comparison of values of $-pK_2$ calculated from (57) at $\theta_2 = 0$ (where (57) is essentially exact) with published measurements of $-pK_{Mg}^0$, the 'intrinsic' Mg^{2+} binding constant extrapolated to $\theta_{Mg} = 0$. The experimental values in the table are not necessarily the numerical values given for the quantity called ' K ' in the cited source; rather, the experimental values in the table conform to the definition of K given in (30), extrapolated to $\theta_{Mg} = 0$. The agreement is quite good, especially if it is remembered that half a pK unit corresponds to a free energy discrepancy of only 0.7 kcal at 25 °C.

In Fig. 1 the Mg^{2+} binding data of Clement *et al.* (1973) at 10^{-3} M ionic strength clearly show the predicted upward curvature of a Scatchard plot. In Fig. 2 data from a variety of sources have all been 'normalized' to 0.1 M ionic strength by applying the theoretical curve

TABLE 5. *Comparison of theoretical and experimental values of K_{Mg}^0*

c_1 (M)	$\log K_{Mg}^0$ (th)	$\log K_{Mg}^0$ (exp)
0.2	1.8	2.2†
0.2	1.8	1.8‡
0.1	2.4	2.6§
0.05	3.0	3.3
0.01	4.4	4.9§
0.001	6.4	5.8¶

† From Skerjanc & Strauss (1968); $M^+ = (CH_3)_4N^+$.

‡ From Zubay & Doty (1958); $M^+ = Na^+$.

§ From Archer, Craney & Krakauer (1972); $M^+ = Na^+$, 'DNA' = poly A·poly U.

|| From Shapiro, Stannard & Felsenfeld (1969); for Ca^{2+} , $M^+ = (CH_3)_4N^+$.

¶ From Clement, Sturm & Daune (1973); $M^+ = Na^+$.

of Fig. 3 to the original data (ranging in ionic strengths from 10^{-3} to 0.2 M). The resulting comparison shows that the ionic strength dependence of K_2 is about right (otherwise, many of the points would lie off the graph) and that the dependence on θ_2 is also given reasonably well. In Fig. 3 Krakauer's curve-fitting expression (1974) for poly A·poly U was used to obtain the points, which accounts for their regularity. The overall variation of slope is about the same for both theory and data points; the detailed curvature as a function of θ_2 cannot be tested on the basis of currently available data. Equation (47) predicts that the primary driving force for non-specific binding of divalent metal ions is the local counterion diffusion potential. If $c_1 \approx c_1^{loc}$, the potential disappears, and binding becomes very weak. Since $c_1^{loc} \approx 1$ M for DNA, the precise prediction is that the association constant for Mg or Ca ion binding should still be appreciable at 0.1 M but should fall to very low values at ionic strengths in the neighbourhood of 1 M. Fig. 2 indicates that K_2 is about 10^2 at 0.1 M. Data are not available for ionic strengths higher than 0.2 M, for which K_{Mg} was measured to be 59 (Zubay & Doty, 1958), but the theoretical prediction

$$d \log K_2 / d \log c_1 = -2,$$

which is more or less in line with observation, suggests strongly that measured values of K_2 at 1 M will be found to be of order unity, if indeed such small values can be measured at all.

3. *Conclusion.* From the reasonably successful comparison of theory and experiment for Mg^{2+} binding I conclude that the molecular model on which the theory is based may be quite realistic; in particular, Mg^{2+}

(and probably many other divalent metal cations) binds to polynucleotides in the delocalized mode. The immediate environment of a bound Mg^{2+} ion is very nearly the same as in free solution; the bound state is characterized by essentially complete hydration and free translational and rotational mobility. In view of the independent evidence outlined in Section II consistent with the same conclusion, the case for delocalized binding of Mg^{2+} is a strong one.

E. Binding of small cations with complex structure

1. *Introduction.* It is reasonable to wonder how well equation (29) may work for cationic substances with relatively small values of N , but with 'structure', typically due to discretely protonated nitrogen atoms on the same molecule. One would certainly worry about the naïve use of (29) for any but the lowest values of θ_N , since excluded volume effects among bound cations have not been considered. However, the few binding data that do exist are fortunately concerned with the 'intrinsic' binding at $\theta_N = 0$. For DNA substitution of the values $\xi = 4.2$, $V_P = 646 \text{ cm}^3$ from Table 1 and $\theta_N = 0$ into (29) yields

$$\log K_N = -0.62 + 0.51N - N \log c_1, \quad (58)$$

and the equation may be easily compared with available data, provided the latter are used in a form consistent with the definition, (30), of K_N .

2. *Oligolysine.* Record, Lohman & de Haseth (1976) have reanalysed the data of Latt & Sober (1967) for the binding of a series of oligolysines with charge state $+N$, $N = 3$ to 8. The experimental results in NaCl could be adequately represented by

$$\log K_N = (-0.62 + 1.46) + (0.51 - 0.68)N - 0.9N \log c_1, \quad (59)$$

where the first two terms have been written as sums to facilitate comparison with equation (58). The polynucleotide used was poly A · poly U, the structure of which is close enough to DNA (see Table 1) to allow direct comparison of (59) with (58). (The reader, if worried by this statement, may change the numerical values -0.62 and 0.51 in (58) and (59) to -0.66 and 0.55 , respectively, to correspond exactly to the specification for poly A · poly U in Table 1.) The form of (59) is correctly predicted by (58). $\log K_N$ is linear in both N and $\log c_1$. As a function of N it has a non-zero intercept that is independent of c_1 and a slope that varies linearly with $\log c_1$. As a function of $\log c_1$ both the intercept and slope vary linearly with N .

Interpretation of the numerical value of the intercept at $N = 0$ is made difficult by the presence of an extra lysine bearing an ϵ -DNP group on each member of the series used in the binding experiments. Equation (58) reflects only ionic and ideal mixing effects; the non-zero intercept is due to the translational entropy of the oligolysine molecule, as a unit, within the free volume that defines the bound state. As suggested by Latt & Sober (1967), the DNP group may contribute significantly to the binding, and its contribution would then show up in the intercept, providing thereby a satisfactory explanation for the discrepancy of 1.46 pK units between measured and predicted intercepts. It should be stressed, however, that observation of a non-zero intercept does not imply by itself a specific interaction independent of the ionic contribution, since (58) possesses a non-zero intercept but is derived exclusively from nonspecific ionic theory.

The linear dependence on N originates theoretically from the difference in ideal entropies of mixing of N M^+ ions in the condensed state relative to bulk solution (that is, from the counterion diffusion potential), since N condensed univalent cations diffuse into bulk solution as one N -valent cation binds. There is reasonable agreement between the theoretical slope $-N$ as a function of $\log c_1$ and the measured result $-0.9N$.

The concentration-independent second term of (59) is 0.68 pK units lower, on an electrostatically equivalent basis (i.e. divide by N), than the corresponding theoretical term in (58). That is, per equivalent bound, this contribution to the binding free energy is observed to be 0.93 kcal less than predicted (at 25 °C). I shall discuss below the possible significance of this fact.

Unpublished data of Lohman and Record for the binding of pentalysine (charge state $N = 5$) to DNA at pH 6 can be represented by (Record, personal communication)

$$\log K_5 = -0.62 + (0.51 - 0.57)(5) - 5.3 \log c_1, \quad (60)$$

where, again, the form facilitates comparison with (58). The observed slope -5.3 of the $\log c_1$ dependence compares favourably with the predicted value -5 from (58). Having, in writing (60), simply *assumed* that the form of (58) applies with the value -0.62 as the correct result of extrapolation of $\log K_N$ to $N = 0$ (the pentalysine used by Lohman and Record does not have the extra ϵ -DNP lys group of the Latt-Sober compounds), I now note that the apparent discrepancy in the

second term is 0.57 pK units per equivalent; at room temperature, the observed binding free energy corresponding to this term is 0.78 kcal/(bound equiv.) less than predicted, a value quite similar to that obtained above for the Latt–Sober series. See Record *et al.* (1978) for a discussion of these data from another point of view.

3. *Spermine*. Spermine is a polyamine with charge state $N = 4$. For the ionic strength $c_1 = 0.15$ M-NaCl, an observed value of K_4 for binding to DNA may easily be extracted from the data of Shapiro *et al.* (1969*b*). The value is $\log K_4 = 2.89$, or

$$\log K_4 = -0.62 + (0.51 - 0.46)(4) - (4) \log(0.15). \quad (61)$$

Comparison with (58) yields an apparent discrepancy in the second term of 0.46 pK units; the observed binding free energy is 0.63 kcal/(bound equiv.) less than predicted at 25 °C.

4. *Lys-trp-lys*. Data for the binding of this tripeptide (charge state $N = 2$) to poly A are available (Brun, Toulmé & Hélène, 1975) and may be represented by (Record, Lohman & de Haseth, 1976)[†]

$$\log K_2 = -1.8 - 2.4 \log c_1. \quad (62)$$

Parameters for poly A are listed in Table 1; the corresponding numerical form of the theoretical equation (29) at $\theta_2 = 0$ is

$$\log K_N = -0.23 - 0.02N - N \log c_1. \quad (63)$$

Accordingly, the empirical equation (62) can be recast into the form

$$\log K_2 = -0.23 - (0.02 + 0.77)(2) - 2.4 \log c_1. \quad (64)$$

The observed coefficient -2.4 of the $\log c_1$ dependence is in reasonable agreement with the value -2 predicted from (63). The discrepancy of 0.77 pK units in the second term corresponds to an observed binding free energy lower than predicted by the amount 1.0 kcal/(bound equiv.) at 25 °C. Record *et al.* (1978) have also analysed these data within the context of their binding theory.

5. *Discussion and conclusion*. It is striking that substances with charge states as diverse as lys-trp-lys and octalysine and with structures as distinct as spermine and oligolysine should all have observed binding free energies lower than predicted by 0.6–1.0 kcal/bound equiv. It

[†] Equation (62) represents the stability constant of the complex denoted as 'A' by Brun *et al.* and conceived by them as not involving interaction of the trp residue with the adenine bases. The analysis given here makes no contact with the question of whether or not A is in dynamic equilibrium with a stacked complex B.

should first be realized that, whatever the biological importance of small binding energies, the molecular theoretician cannot help but be pleased by such small discrepancies, of the order of random thermal energy, between observation and *a priori* calculation. It is fair to conclude that the model of delocalized binding on which the theory rests may be realistic for these cationic substances. The observation that predicted binding constants *overestimate* measured values strengthens this conclusion, since localization of the cation in potential energy wells on the polynucleotide could be expected to require sources of binding free energy *in addition to* those considered in the model. The difference between observation and an *underestimated* prediction could have been consistently attributed to localization; the overestimate actually found does not encourage this line of reasoning. At the present time, therefore, I believe that interpretations of the binding of these substances in terms of ion-pair formation, stabilized by hydrogen bonds, are not warranted. Binding to the bases has been ruled out for spermine and tetralysine by the differential dialysis measurements of Shapiro, Stannard & Felsenfeld (1969).

The molecular origin of the discrepancy between theory and experiment could be a conformational change induced in the cation while in the bound state. Schematically, one may speculate as follows. When free in solution octalysine tends to project its charged lysine residues on alternate sides of the molecule. In this conformation, not all eight residues can fit simultaneously into the bound region V_P , which is only 7 Å thick, and octalysine would bind like a cation of valence lower than eight. However, if all lysine residues are brought to the same side of the molecule, octalysine can bind like an octavalent cation. Since octavalent cations preferentially displace cations of lower valence (for the same reason that Mg^{2+} displaces Na^+ ions), the latter conformer will be the bound form. The process of binding can be thermodynamically described as the sequential transformation from a '*trans*' to a '*cis*' form in solution followed by binding of the *cis* form. The first step involves bringing like charges closer together (and possibly other strains) and hence costs free energy. The observed binding constant is therefore smaller than what would be expected of a structureless octavalent cation. The structure of spermine suggests less conformational energy expenditure per equivalent than for oligolysine, and the corresponding discrepancy between theory and experiment is in fact the least of all the substances considered. The bulky tryptophan residue may impose

more severe restrictions on the spatial arrangement of the lysine residues in lys-trp-lys; the corresponding discrepancy is indeed the greatest of those considered.

F. Binding of high-molecular-weight cationic substances

1. *Introduction.* Complexes of DNA with a surprisingly large variety of cationic substances are observed to dissociate when the ionic strength is raised to about 1 M. The simplest example, the Mg/DNA complex, has already been discussed in Section IV D. Additionally, the empirical formulas in the preceding section for the binding of oligolysine show association constants that drop off to values of the order unity at 1 M ionic strength. The phenomenon is not restricted to small cations; polylysine/DNA complexes (Leng & Felsenfeld, 1966; Shapiro, Leng & Felsenfeld, 1969) and nucleohistones in chromatin (Weintraub, Palter & van Lente, 1975) dissociate at about 1 M as well. An ionic strength of 0.6 M is sufficient to elute DNA from DEAE columns (Hirsh & Schleif, 1976). Extrapolation of binding data for the complex formed from *lac* repressor and non-operator DNA (de Haseth, Lohman & Record, 1977; Record *et al.* 1978) suggests that the binding constant is of order unity at about 0.5 M ionic strength even though it is of order 10^6 at 0.1 M.

It is likely that such close phenomenological similarity has its molecular origin in structural features possessed in common by all these systems. In fact the above behaviour would be *predicted* under the assumption that all these binding processes are predominately driven by the counterion diffusion potential, the local tendency of counterions to diffuse into free solution from a region of high concentration, which for DNA has the value 1.2 M.

In the absence of clear evidence to the contrary, therefore, it does not seem necessary to invoke a specific, geometrically complementary, structure for each of the complexes, nor is it necessary to conceive of cationic groups as having entered into hydrogen-bonded ion pairs with the phosphate oxygen atoms. The dominant driving force for complexation is brought into play simultaneously with penetration of cationic groups into the free volume V_p containing a locally high concentration of univalent counterions. The resulting complex is 'loose'; the cationic substance (or positively charged region of a protein) is free to slide along the DNA surface with a layer or two of water molecules intervening. Of course, the sliding motion of a protein (a histone, for

example) can be prevented by purely mechanical means if the DNA helix is wrapped around it.

In the following paragraphs I shall discuss what appear to be implications of this description for three cases of interest: polylysine-DNA, repressor-DNA, and repressor-operator.

2. *The polylysine/DNA complex.* The possibility that polylysine forms a loose *non-specific* complex with DNA in the sense of the preceding paragraph is supported by at least three lines of evidence. As discussed above, the complex dissociates at about 1 M ionic strength, implying stabilization by realization of the non-specific counterion diffusion potential (tendency of counterions to diffuse into free solution from a *loosely* bound state). By direct calorimetry, Ross & Shapiro (1974) showed that the binding of polylysine to DNA is accompanied by only very small enthalpy changes; it is primarily entropy-driven, as would be true for a process driven by local counterion diffusion. (Ross & Shapiro also determined that the same is true for the binding of Mg^{2+} , spermine, and tetralysine.) Finally, Mirzabekov & Melnikova (1974) measured the effect of polylysine binding on methylation of the $N_{(7)}$ atom of guanine in the major groove of DNA and $N_{(3)}$ of adenine in the minor groove. Polylysine had no effect on the methylation reaction in either groove, in contrast to significant interference effects found for a variety of other binding substances. The conclusion suggested by the authors was that polylysine does not make contacts in either groove when bound to DNA. Indeed, it should be asked whether their data do not definitively rule out models which have the entire polylysine backbone buried in a groove with the cationic residues 'sticking out' and attached to the phosphates.

The indisputable fact remains that polylysine can quantitatively precipitate A·T-rich DNA from a mixture of A·T-rich and G·C-rich DNA (Shapiro, Leng & Felsenfeld, 1969). It may be, however, that this phenomenon has its molecular basis entirely in the base-specific properties of DNA itself, and not in any specific properties of the complex. If A·T-rich DNA were inherently more flexible than G·C-rich DNA, then even the loose, nonspecific binding of polylysine to the former could more easily induce folding of the double helix in the vicinity of the bound polylysine molecule. In effect, then, polylysine may bind to a more completely 'doubled over' region of DNA if the latter is rich in A·T base pairs, hence to a region of somewhat higher charge density than would be available in G·C-rich DNA. Even a 5'

differential increase in charge density would be enormously magnified, first because, according to (16), the local concentration c_1^{loc} of univalent counterions would be relatively increased by the factor $(1.05)^2 = 1.10$, and then because (47) implies that the ratio of affinities for a 50-residue polylysine bound to A·T-rich relative to G·C-rich DNA would be $(1.10)^{50}$, or two orders of magnitude. Even a 1% effect in the charge densities translates into an affinity ratio of 3 for a 50-residue polylysine and 7 for 100 residues. Concerning the hypothesis that A·T-rich DNA may be less stiff than its G·C-rich counterpart, there is no evidence one way or the other as far as I know; nevertheless, in view of the detailed analysis in Section VI of electrostatic effects on DNA flexibility, it may be a reasonable conjecture.

3. *The complex of lac repressor with non-operator DNA.* As mentioned in paragraph 1 of this section, the affinity of *lac* repressor for non-operator DNA extrapolates to very low values at ionic strengths of the order of 1 M, a behaviour consistent with a loose and mobile complex. In addition, no volume change on binding could be detected by Revzin & von Hippel (1977), the likely implication being that the cationic groups on repressor (the binding region of repressor, for both operator and non-operator DNA, is known to be cationic), do not approach DNA sufficiently closely to release water of hydration from either repressor or DNA.

Methylation protection experiments (Kolchinsky *et al.* 1976) indicate, in agreement with other evidence (Lin & Riggs, 1972; Richmond & Steitz, 1976), that repressor makes no contacts in the major groove of non-operator DNA. Repressor does protect the minor groove against methylation, but the measure of protection is small, about 15%, a figure that can be contrasted to the 78% effect obtained with ethidium bromide.

These considerations are all consistent with a counterion-potentiated complex (de Haseth *et al.* 1977) perturbed by interactions in the minor groove that are too weak to prevent either substantial repressor mobility when bound or complete dissociation when the ionic strength is raised. It is reasonable to speculate that the weak interactions may involve contacts between cationic residues in the binding region of repressor with thymine O₍₂₎ atoms located in the minor groove (see Section III F).

4. *Repressor/operator complex.* The repressor/operator complex is specific. Under physiological conditions its binding affinity is some eight orders of magnitude higher than the affinity of repressor for non-operator DNA (Record, de Haseth & Lohman, 1977; Revzin &

von Hippel, 1977). Its binding constant does not extrapolate to a small value at 1 M salt (Record *et al.* 1977). However, the same region of repressor that is responsible for its DNA binding capability is also necessary for binding to the operator segment (Bourgeois & Pfahl, 1976). Moreover, the ionic strength dependencies of the binding to operator and to non-specific DNA, although significantly different, are similar both in form and magnitude (Record *et al.* 1977). It can therefore be concluded (Record *et al.* 1977) that the stability conferred on the repressor/DNA complex by the counterion diffusion potential is largely transferred to the repressor/operator complex, and that the enhanced stability of the latter is due to superposition on the counterion potential of a rather strong interaction in the grooves.

Methylation experiments (Gilbert, Maxam & Mirzabekov, 1975) show that both major and minor grooves on the operator are protected when repressor is bound. Taking into account their result that repressor interacts weakly with only the minor groove in non-operator DNA, Kolchinsky *et al.* (1976) have proposed that when repressor initially binds to a non-operator region of operator-containing DNA, it proceeds to wander around, poking into the minor groove until it roughly recognizes the operator, whereupon it clamps down firmly on the operator, surrounding it and interacting strongly in both grooves. To supplement this attractive picture I offer a conjecture which may be extreme but that is nonetheless consistent with available data. If, as suggested in paragraph 3, the occasional probing by repressor of the non-operator minor groove is accomplished by means of weak contacts of one or two cationic residues with thymine $O_{(2)}$ atoms, it is then possible that, upon encountering a sterically specific sequence of thymine $O_{(2)}$ atoms in the minor groove of the operator, the repressor falls into a deep potential well created by entropic release of one or two water molecules from each of several 'partially ionic pairs' formed from cationic residues and thymine $O_{(2)}$'s. The repressor could be pictured as swinging around to cover the major groove, without actually forming stabilizing contacts there, although that could happen also. This conjecture is consistent with the known facts that the 'extra' stability of the repressor/operator complex, over and above that of the non-operator DNA/repressor complex, is also entropic (Riggs, Bourgeois & Cohn, 1970), and that under conditions conducive to dehydration thymine and uracil $O_{(2)}$ atoms are substantially anionic in character (Section IIIF).

V. IONIC EFFECTS ON THE DNA HELIX-COIL TRANSITION

A. *Electrostatic free energy and entropy of transition*

The coil form of DNA (separated strands) has a lower charge density than the native duplex helical form. It is expected, therefore, that the electrostatic free energy in the coil phase is lower than that in the helix phase. To proceed quantitatively, define $\Delta\bar{g}$ as the difference 'coil minus helix' of the free energies per mole phosphate in units of RT , and let $\Delta\bar{g}^0$ refer to the non-electrostatic contribution. The prescription given in Section IIC for calculating the free energy of a polyelectrolyte solution characterized by any value of b (axial charge spacing) is now applied to both coil and helix forms of DNA, and the difference is taken. The result for DNA in an aqueous solution of excess 1:1 salt is

$$\Delta(\bar{g} - \bar{g}^0) = \Delta[(1 - \xi^{-1}) \ln(c_1^{100}/c_1) - \xi^{-1} \ln(\kappa b)], \quad (65)$$

where all quantities have been defined in Section II. The numerical result may be obtained by reference to Table 1 and equation (5),

$$\Delta\bar{g} = \Delta\bar{g}^0 - 1.11 + 0.36 \log c_1, \quad (66)$$

where the coefficient of $\log c_1$ is the numerical value of $\frac{1}{2}(\Delta\xi^{-1}) \ln 10$. The ionic contribution to the enthalpy change, which is contained in the slight temperature dependence of the product of absolute temperature and bulk dielectric constant of water, is less than 0.1 kcal per mole phosphate and may be neglected for practical purposes. Equation (66) implies therefore that the ionic contribution to the entropy change is about 3 entropy units per mole phosphate for ionic strengths in the neighbourhood of 0.1 M. For comparison, the total entropy change from all sources is about 11 or 12 entropy units per mole base.

B. *Dependence of T_m on ionic strength*

The dependence of T_m on ionic strength is implicit in the defining relation

$$\Delta\bar{g}(T_m, c_1) = 0. \quad (67)$$

Application of the Gibbs-Helmholtz identity to both sides of this equation, together with equation (65), gives (Manning, 1972b)

$$dT_m/d \log c_1 = \frac{1}{2}(\Delta\xi^{-1})(2.30RT_m^2/\Delta h^0), \quad (68)$$

where Δh^0 is the enthalpy change *per mole phosphate* at the melting temperature. The value of $\Delta\xi^{-1}$ from Table 1 is 0.32. It is an empirical fact that the quantity $2.30RT_m^2/\Delta h^0$ is nearly independent of T_m . Values

determined from three separate calorimetric studies are 115° , 129° and 156° (Record, Woodbury & Lohman, 1976). The corresponding theoretical values of $dT_m/d \log c_1$ from equation (68) are 18° , 21° and 25° respectively. A recent experimental value of $dT_m/d \log c_1$ is 19.5° (Record, 1975). Agreement between theory and experiment is close; the 'discrepancy' noted in a previous publication (Manning, 1972*b*) was due to use of an incorrect value for b for the coil form.

A few words should be devoted at this point to the appropriate values of b . For the duplex helix the value $1.7 \text{ \AA}$ given in Table 1 is merely the familiar $3.4 \text{ \AA}$ step height of the B form divided by the two phosphate groups per step. For the separated strand the value $b = 4.0 \text{ \AA}$ in Table 1 was chosen as that which provides the best average agreement of (68) with measurement (Record, Woodbury & Lohman, 1976; Manning, 1976*a*). However, closely similar values of b for the coil have also been obtained by independent means (Manning, 1976*a*). The value $3.8 \text{ \AA}$ was determined by low-angle X-ray scattering on poly C in solution at elevated temperature (Gulik, Inoue & Luzzati, 1970). An identical value was deduced from an *a priori* conformational energy analysis, assuming equal proportions of C(3')-endo and C(2')-endo residues and the absence of all-*trans* arrangements of the phosphodiester link (Olson & Manning, 1976). Therefore equation (68) can either be viewed as a means of converting calorimetric and melting temperature measurements into a numerical value of the axial charge spacing for a single strand, or as an independent theoretical equation which accurately predicts the ionic strength dependence of T_m from calorimetric data and known values of the charge spacings of helix and coil.

C. Role of the counterion diffusion potential

The coil form of DNA (separated strands) has a lower charge density than the native duplex helical form. Therefore the structural transition native $\rightarrow$ coil is a mechanism for relief of phosphate-phosphate repulsive stress. At temperatures greater than T_m this mechanism for compensation of phosphate repulsion becomes available to the system as an alternative to counterion binding. Therefore, in accordance with the discussion at the end of Section II C, the potential for local counterion diffusion is realized and accompanies the spontaneous transition as the ionic contribution to the overall driving force, identifiable quantitatively as the first term on the r.h.s. of equation (65).

The second term in (65) is also electrostatic but acts in the opposite

direction to stabilize the helix. Its physical origin is in the fact that the net charge per phosphate, $-\xi^{-1}q$, is lower for the helix than for the coil, so that screened electrostatic repulsion among pairs of phosphates is *greater* for the coil form. The relation $(\kappa b)^2 = e^{-1}(c_1/c_1^{\text{lo}})$ mentioned in IIC may be used to show numerically that this 'counter-driving force' is considerably smaller than the counterion diffusion term. Interestingly, however, the magnitude of the ionic-strength dependence (the $\log c_1$ term) contributed by this effect is fully 50% as large as that from counterion diffusion.

A qualitative prediction following from this discussion is that the ionic-strength dependence of T_m should level off as c_1 approaches c_1^{lo} , or about 1 M. Precisely this behaviour is in fact commonly observed (Schildkraut & Lifson, 1965). The related consequences of ionic strengths of about 1 M for multivalent cation binding should be recalled (see Section IVF, 1).

D. Effect of strong electric fields on T_m

The dissociation-field theory described in Section IIIE may be used to calculate the effect of an electric field on the stability of the DNA double helix (Manning, 1977c). Let $\bar{g}(0)$ be the free energy, minimized as in Section IIC, of a polyelectrolyte solution in the absence of an electric field (in units of RT per polyelectrolyte equivalent); the corresponding quantity in the presence of a field E is designated by $\bar{g}(E)$. The explicit free energy calculation gives, to first order in E ,

$$\bar{g}(E) - \bar{g}(0) = (1 - \xi^{-1}) \bar{E}, \quad (69)$$

where $\bar{E}$ has been defined by equation (23) and all counterions are assumed univalent. The physical meaning of this equation is transparent; the number of condensed counterions is $1 - \xi^{-1}$ per polyion charge, and each of these ions has the additional energy $\bar{E}$ (in units of kT) when the field is applied. The actual dissociation of some of the condensed counterions enters only as a negligible second-order effect. It is understood that the field is 'pulsed' for a time sufficiently short to eliminate orientational effects.

If this relation is applied both to double helical DNA and to its separated strands,

$$\Delta \bar{g}(E) - \Delta \bar{g}(0) = \Delta[(1 - \xi^{-1}) \bar{E}]. \quad (70)$$

The quantity $\bar{E}$ is slightly different for single strands and double helix

(since the parameter a in equation (23) depends on ξ), but the dominant effect is simply that $1 - \xi^{-1}$ is larger for the double helix relative to separated strands (the right-hand side of equation (70) is negative). Intact DNA binds the greater number of cations and hence suffers the greater increase of energy when the field is turned on ($\Delta\bar{g}(E) < \Delta\bar{g}(0)$). The free energy of the double helix becomes relatively higher, and the tendency for strand separation is enhanced. The T_m is lowered by the field. Upon insertion of the appropriate numerical values equation (70) becomes

$$\Delta\bar{g}(E) - \Delta\bar{g}(0) = -0.94 \times 10^{-3} E, \quad (71)$$

if E is in kV/cm.

Now if strand separation is actually to occur, $\Delta\bar{g}(E)$ must be reduced by the field to the value zero. Field-induced strand separation is a *threshold* phenomenon that does not occur for fields below a critical, or threshold, value. Let E_{th} denote the threshold field. Then, from (71)

$$E_{th} = 1060 \Delta\bar{g}(0). \quad (72)$$

It follows immediately from equation (66) that E_{th} is linear in $\log c_1$ with

$$dE_{th}/d \log c_1 = 350 \text{ kV/cm}. \quad (73)$$

Field-induced melting has not been unambiguously observed. The predicted effect is small; experiments would have to be done at an ionic strength such that the T_m in the absence of the field is only a few degrees higher than the temperature of the apparatus. Melting might then be observed at a field of order 50 kV/cm (Manning, 1977*c*). At significantly higher ionic strengths (higher T_m) equation (73) suggests that the highest attainable fields might not be sufficient to induce melting (i.e. to lower the T_m to the ambient temperature). Detailed measurements on the non-cooperative unstacking of poly C, however, have revealed a threshold field that varies linearly with the logarithm of the ionic strength (Pörschke, 1976*b*). It is not clear at the moment whether it is easier to extend the theory to non-cooperative melting or to do the experiment for DNA.

E. *Effect of pH on T_m ; average axial phosphate spacing of single strand DNA as a function of pH*

Titration of the bases with acid or alkali affects the polyelectrolyte behaviour only by changing the value of ξ (Record, Woodbury & Lohman, 1976). Heretofore, ξ was considered as being built up by the

negative phosphate charges only; the bases were tacitly assumed to be untitrated, and (68), for example, applies only to the pH band about pH 7 for which the bases are in fact uncharged, at least if Table 1 is to be used for the numerical values of ξ . Protonation of bases decreases the value of ξ ; deprotonation augments the negative phosphate charge, and consequently ξ is increased.

For a given pH, let the extent of titration per nucleotide be k_h and k_c for duplex DNA and a coiled single strand, respectively. Then (Record, Woodbury & Lohman, 1976)

$$\xi'_h = \xi_h(1 \mp k_h), \quad (74)$$

where ξ'_h is the value of ξ to be used in polyelectrolyte calculations, ξ_h is its value based only on the phosphate charges (equal to 4.2), and the upper and lower sign refer, respectively, to acid and alkaline titration. An analogous expression holds for the separated strands. The number of condensed M^+ cations per nucleotide is $1 - \xi'^{-1}_h$, while the screening free energy per mole nucleotide given by equation (8) for the neutral pH case with $N = 1$ turns out to be unmodified because of the distinction that must now be made between moles of nucleotides and polyion equivalents (Record, Woodbury & Lohman, 1976). The result for the ionic strength dependence of T_m , analogous to equation (68), is

$$dT_m/d \log c_1 = (2.30RT_m^2/\Delta h^0) (\frac{1}{2}\Delta\xi^{-1} \pm \Delta k). \quad (75)$$

Note that if the bases are untitrated ($k_h = k_c = 0$), or if there is no *differential* titration ($\Delta k = 0$), then equation (75) reduces to equation (68). The situation is somewhat more complicated than indicated by equation (75) since acid titration can in practice lower ξ' to the extent that it becomes less than unity, so that no M^+ cations condense (recall that the charge fraction is ξ'^{-1}) and other expressions apply for the free energy (Manning, 1969). The details are presented in full by Record, Woodbury & Lohman (1976).

The extents of titration k_c and k_h are experimentally accessible at each pH, as is $dT_m/d \log c_1$ and Δh^0 . It may be assumed that $\xi_h = 4.2$ regardless of pH because of the double helical rigidity. However, it is possible that the average axial nucleotide spacing on the single strand may increase with the total charge placed on the chain. Hence, ξ_o is regarded as the sole adjustable parameter in equation (75) and the values of b_o in Table 6 are those required to bring (75) into agreement with data (Record, Woodbury & Lohman, 1976). It is seen from the table that the

TABLE 6. *Variation of single-strand nucleotide spacing (b_e) with pH†*

pH	ξ_e	b_e (Å)
11.3	1.14	6.3
11.0	1.19	6.0
10.8	1.27	5.7
9.0	1.49	4.8
7.0	1.72	4.1
4.8	2.50	2.9
4.5	2.78	2.6
3.7	4.00	1.8
3.2	4.17	1.7

† Reproduced with permission from Record, Woodbury & Lohman (1976).

axial nucleotide spacing (contour length divided by number of bases) does indeed increase with pH, reaching the value characteristic of a fully extended chain at the highest pH. Apparently the all-*trans* conformation, which may be excluded at neutral pH (Olson & Manning, 1976) becomes dominant upon sufficient deprotonation of the bases with consequent strong close-range electrostatic repulsion. Finally, the value at the lowest pH suggests that the single strand has 'folded over' on itself and adopted a spacing similar to that of the double helix (recall that $b_h = 1.7$ Å and $\xi_h = 4.2$).

F. Ionic effects of strandedness on T_m

The four transitions of the (poly A, poly U) system provide a nice test of the theory (Record, Woodbury & Lohman, 1976). Only two of the four transitions are thermodynamically independent and may be chosen as

- (1) poly A · poly U $\rightarrow$ poly A + poly U,
- (2) poly A · 2 poly U $\rightarrow$ poly A + 2 poly U.

It is easy to show (Record, Woodbury & Lohman, 1976; de Marky, unpublished) that the corresponding theoretical equations for

$$dT_m/d \log c_1$$

have the same form as (68),

$$dT_m/d \log c_1 = \frac{1}{2}(\Delta_i \xi^{-1}) (2.30RT_m^2/\Delta h^0), \quad (76)$$

but with $\Delta_i \xi^{-1}$ given by the number-averaged differences,

$$\Delta_1 \xi^{-1} = \frac{1}{2}\xi_A^{-1} + \frac{1}{2}\xi_U^{-1} - \xi_{A \cdot U}^{-1} \quad (77)$$

$$\Delta_2 \xi^{-1} = \frac{1}{3}\xi_A^{-1} + \frac{2}{3}\xi_U^{-1} + \xi_{A \cdot 2U}^{-1}, \quad (78)$$

where the subscripts '1' and '2' refer to the transition number. If it is assumed that $\xi_A = \xi_U$, then $\Delta_i \xi^{-1}$ collapses to the quantity $\Delta \xi^{-1}$ of equation (68). The values of $\xi_{A,U}$ and $\xi_{A,2U}$ are known from X-ray data: $\xi_{A,U} = 4.6$, $\xi_{A,2U} = 6.8$ (Record, Woodbury & Lohman, 1976). The required calorimetric and T_m data are available (Krakauer & Sturtevant, 1968) and have been tabulated in convenient form by Record, Woodbury & Lohman (1976).

Let us assume first that $\xi_A = \xi_U = 1.9$, the numerical value for unstacked (high-temperature) poly C (see Section VB). The values of $dT_m/d \log [Na^+]$ predicted from (76) are then 20° for transition (1) and 24° for transition (2); the observed values are 19.6° and 26° respectively. For the K^+ salt the $\Delta h^0/T_m^2$ data are slightly different. Predicted values of $dT_m/d \log [K^+]$ are 21° and 28° for transitions (1) and (2), respectively, while the corresponding measured values are 17.4° and 27° .

Equation (76) may be used in another way (Record, Woodbury & Lohman, 1976). Written for both transitions (1) and (2), two equations in the two unknowns ξ_A and ξ_U are obtained, all other quantities being given from the available data. The values $\xi_A = 2.3$ and $\xi_U = 1.6$ are obtained from the Na^+ data; the K^+ data yield $\xi_A = 3.1$ and $\xi_U = 1.6$. It is not unreasonable that the axial phosphate spacing should be greater for unstacked poly U than for poly A, which is partially stacked even at typical melting temperatures. The axial spacing b for poly U corresponding to $\xi = 1.6$ (obtained in both Na^+ and K^+ salts) is $4.5 \text{ \AA}$, which could be slightly high (Manning, 1976*a*). The axial spacings for poly A corresponding to the ξ values 2.3 (Na^+) and 3.1 (K^+) are $3.1 \text{ \AA}$ and $2.3 \text{ \AA}$. The former value is essentially the same as that which characterizes the fully stacked low-temperature form of poly A (Stannard & Felsenfeld, 1975), but the latter value is much too small to be rationalized. Using a similar method, Pless & Ts'ao (1977) have determined the axial charge spacings for poly dA and dT₈ as 3.3 and $4.9 \text{ \AA}$, respectively.

G. Effect of Mg^{2+} on T_m

Section IVB outlined a method for obtaining the free energy of a polynucleotide solution in which Mg^{2+} ions are present along with excess 1:1 salt. Therefore it is possible to calculate $\Delta \bar{g}$ and T_m for the DNA melting transition under these conditions (deMarky & Manning, in preparation). Theory and measurements are compared in Fig. 6. The

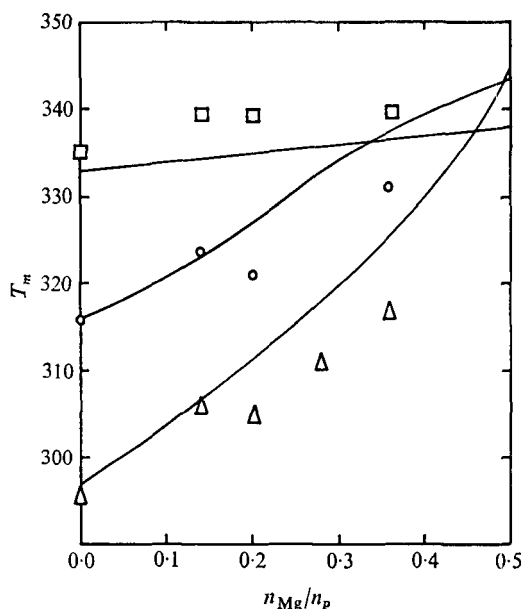


Fig. 6. Melting temperature of DNA as a function of added Mg^{2+} at three different ionic strengths, 10^{-2} , 10^{-3} and 10^{-4} M, from top to bottom at left. The theory (solid curves) has been adjusted to match the data at the single point of zero Mg^{2+} and 10^{-3} M (middle curve at origin). Data from Record (1975).

abscissa in the figure is equivalents of total added Mg^{2+} per mole phosphate, so that the points and curves refer to low binding levels of Mg^{2+} . The data points are taken from Record (1975). The single theoretical point at 10^{-3} M-NaCl and $n_{\text{Mg}} = 0$ is forced to coincide with the measured value (since T_m reflects a non-ionic contribution in addition to ionic effects), but all other points on the curves are determined from *a priori* theory with the sole parameters b_h and b_o , both of which are known independently. It is especially to be noted that the near-independence of T_m on added Mg at 10^{-2} M-NaCl, both as observed and predicted by theory, does not mean that Mg^{2+} ions are not bound at this ionic strength. The theoretical and measured binding constants in Table 5 show that Mg remains strongly bound at 10^{-2} M, at least to the double helix, and theoretical values of K_{Mg} for the single strand are also appreciable (an order of magnitude lower). Therefore the invariance of T_m at this ionic strength reflects a delicate differential balance between Mg^{2+} binding to the two DNA forms, and the success-

ful prediction correspondingly suggests a high degree of theoretical accuracy.

One of the more interesting phenomena associated with Mg^{2+} binding is the cross-over of the curves seen at the high end of the added Mg^{2+} range in Fig. 6. The data do exhibit such a cross-over (Record, 1975; Dove & Davidson, 1962), but at somewhat higher values of added Mg^{2+} . Beyond the region of cross-over addition of 1:1 salt (increased ionic strength) results in destabilization of the double helix for a fixed value of total added Mg^{2+} . The explanation of this phenomenon (Manning, 1972*c*; deMarky & Manning, 1975) is particularly simple for levels of added Mg^{2+} sufficiently high that its binding is essentially saturated (polynucleotide charge fraction equal to $\frac{1}{2}\xi^{-1}$) for both double helix and single strand over a range of ionic strengths. In Section C it was pointed out that melting in the absence of Mg^{2+} is accompanied by release of condensed M^+ ions (increased entropy) and an increase of repulsive electrostatic free energy. The former effect dominates, and, since it destabilizes the double helix but decreases in magnitude as the ionic strength increases, the double helix becomes more stable at higher ionic strengths. At saturating levels of Mg^{2+} , however, no M^+ cations are condensed, having been entirely replaced by Mg^{2+} . The melting transition is consequently unaccompanied by release of condensed M^+ , since there are none to be released. The only remaining effect dependent on ionic strength is the increase of electrostatic free energy, which stabilizes the double helix but decreases in magnitude with increasing ionic strength. Therefore an increased ionic strength destabilizes the double helix.

The effect of Mg^{2+} on T_m has also been discussed by Krakauer (1974) and by Record *et al.* (1978).

H. Ionic effects on the renaturation rate

The rate of association of separated but complementary polynucleotide strands to form double helices is strongly dependent on ionic strength (Wetmur, 1976). Measured values of the apparent second-order rate constant k_2 (Ross & Sturtevant, 1960; Studier, 1969) can be fitted by the empirical relation (Manning, 1976*b*)

$$d \log k_2 / d \log c_1 \approx 3.6. \quad (79)$$

The theory (Manning, 1976*b*) is based on the assumption that the rate-determining step is a pre-equilibrium between separated strands

and a non-bonded precursor to the nucleated species (Wetmur & Davidson, 1968). The problem then reduces to the calculation of $\Delta\bar{g}$ for the process of bringing together two separated strands, each characterized by the charge-density parameter ξ_c , to form an aligned but non-bonded double-stranded configuration of charge density $2\xi_c$ and of length Q base pairs. The theoretical prediction is then easily demonstrated (Manning, 1976*b*) to be

$$d \log k_2 / d \log c_1 = Q / 2\xi_c. \quad (80)$$

Comparison of (79) and (80) with the value $\xi_c = 1.9$ from Table 1 yields the value $Q = 14$, which is comparable to the persistence length of an unstacked single strand.

The effect of added Mg^{2+} has also been treated, and the observed second-order dependence of the association rate on Mg^{2+} concentration (Felsenfeld & Rich, 1957) has been demonstrated to be consistent with the value of Q given above (Manning, 1976*b*).

I. *The mistake of adopting the Debye-Hückel approximation*

A frequently used procedure to calculate the ionic contributions to the free energy of a closely interacting array of charge centres is pair-wise summation of Debye-Hückel potentials. Although everyone is aware that some sort of approximation is thereby being made, just how bad this approximation really is may not be sufficiently appreciated. The helix-coil transition provides a good example.

Suppose pair-wise summation of Debye-Hückel potentials is performed for an infinitely long, linear array of charges with spacing b , and that the difference is then taken for two values of b . The result, which should immediately be compared with (65), is

$$\Delta(\bar{g} - \bar{g}^0) = -\Delta(\xi \ln \kappa b). \quad (81)$$

The comparison shows clearly that, by ignoring the physical reality of bound counterions and their degrees of freedom, the Debye-Hückel procedure has completely missed the molecular phenomena underlying the transition! Another qualitative error is the direct variation with ξ in equation (81); the correct variation is inverse.

Substitution of numerical values from Table 1 into equation (81) yields

$$\Delta\bar{g} = \Delta\bar{g}^0 - 2.93 + 2.76 \log c_1, \quad (82)$$

which may be compared with (66). Although (82) does give the correct

form of the ionic-strength dependence, the slope is an order of magnitude too high. At 0.1 M and 25 °C the electrostatic free energy (sum of the last two terms) predicted by equation (82) is -3.37 kcal/mole P, about four times the value -0.87 kcal/mole P from the more accurate (66). Similarly, the slope of T_m v. $\log c_1$ is about an order of magnitude too high in the Debye-Hückel approximation.

Within the context of such gross inaccuracy, use of exact coordinates for the charge centres can provide only the illusion of computational sophistication.

VI. ELECTROSTATIC CONTRIBUTION TO DNA STIFFNESS

A. Harrington's measurements of persistence length

Recent measurements (Harrington, 1978) of high-molecular-weight DNA persistence lengths in aqueous NaCl over a range of nearly three magnitudes of ionic strength provide the possibility of new insight into local DNA structure. Harrington's data are portrayed in Fig. 7 in the form of a graph of persistence length a (in ångström units) against $\log c_1$, the logarithm of ionic strength. The measurements cover the range $0.005 \text{ M} \leq c_1 \leq 2.0 \text{ M}$. The solid line with greater numerical slope is the least-squares linear best fit based on the four points of lowest ionic strength. The equation of this line is

$$a = 221 - 559 \log c_1 \quad (\text{based on } c_1 \leq 0.2 \text{ M}). \quad (83)$$

The other straight line in Fig. 7 is the corresponding fit based on the five most dilute concentrations; its equation is

$$a = 317 - 504 \log c_1 \quad (\text{based on } c_1 \leq 0.6 \text{ M}). \quad (84)$$

Although there are not enough points to determine unambiguously whether a is an inherently linear function of $\log c_1$ at the low end of the ionic strength range, the fit is reasonable. Moreover, Fig. 7 is strongly reminiscent of observed plots of T_m v. $\log c_1$, which also begin to level off from steep linearity at ionic strengths between 0.2 and 0.6 M (Schildkraut & Lifson, 1965). As discussed in Section V, this type of dependence is related to the counterion diffusion potential. I have accordingly sought for a conformational change, closely linked to the persistence length, which may be driven by the counterion diffusion potential.

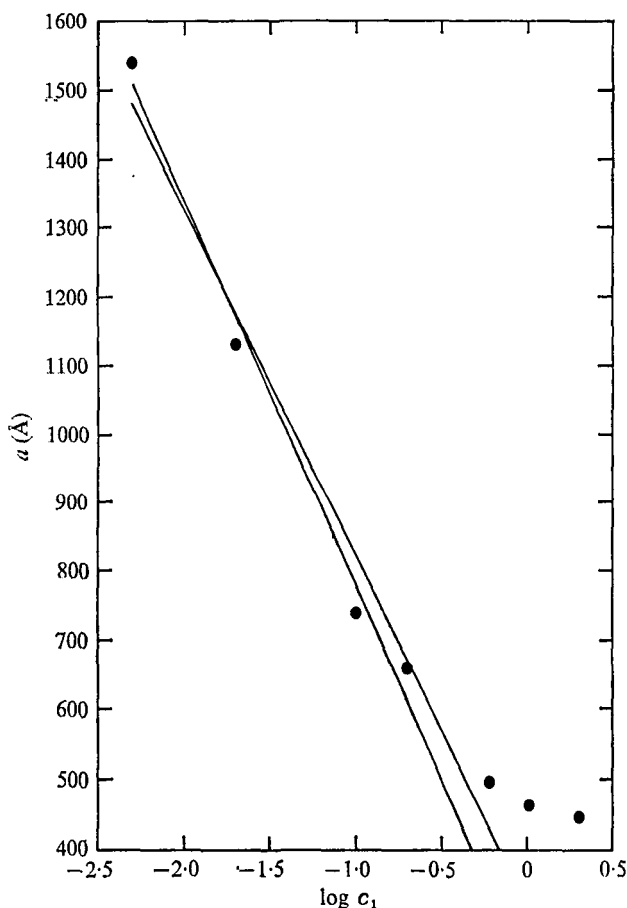


Fig. 7. Harrington's persistence length data (1978) in logarithmic representation. Solid lines are linear regressions as described in text.

B. Polyelectrolyte theory of the persistence length

Suppose that on a scale of length l high-molecular-weight DNA behaves like a rigid rod with elastic bending capability. Then a positive free energy ΔG must accompany the bending of a mole of segments of length l through a small angle θ (see Fig. 8). The persistence length is related to ΔG by the formula (Bloomfield *et al.* 1974)

$$\Delta G = (RTa/2l) \theta^2. \quad (85)$$

It is the spontaneous process bent $\rightarrow$ straight, with negative free energy

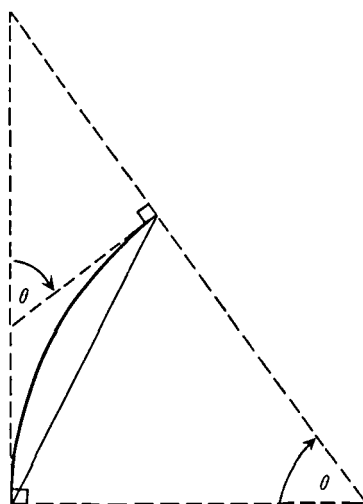


Fig. 8. Geometrical construction used to derive equation (93).

$-\Delta G$, that I presently believe, on the basis of the analysis to follow, to be primarily driven by local counterion diffusion.

With reference to the geometrical construction in Fig. 8, where the heavy circular arc is a segment of DNA of length l bent through an angle θ , let us first note the relation between l , θ , and the end-to-end distance of the arc, denoted by $l(\theta)$ and represented in the figure by the thin solid line:

$$l(\theta) = \sqrt{2} l \theta^{-1} \sqrt{(1 - \cos \theta)}. \quad (86)$$

Expansion of $\cos \theta$ for small angles θ yields

$$\Delta l = -\frac{1}{24} l \theta^2, \quad (87)$$

where

$$\Delta l \equiv l(\theta) - l. \quad (88)$$

When $\theta = 0$, the segment is straight, its end-to-end distance $l(0)$ is the same as its contour length l , and $\Delta l = 0$.

Now denote by n_l the number of phosphates in the segment. This quantity is independent of θ . Define an effective charge spacing $b(\theta)$ of the bent segment by

$$b(\theta) = l(\theta)/n_l. \quad (89)$$

Then, corresponding to equations (87) and (88),

$$\Delta b = b(\theta) - b = -\frac{1}{24} b \theta^2, \quad (90)$$

where $b = l/n_l$, the usual axial charge spacing (1.7 Å from Table 1).

The effective charge spacing $b(\theta)$ in the bent conformation is less than that of the straight conformer. The corresponding formula for $\xi(\theta)$, the charge density parameter generalized from equation (1) as proportional to the reciprocal of $b(\theta)$, is

$$\Delta\xi = \xi(\theta) - \xi = \frac{1}{24}\xi\theta^2. \quad (91)$$

The charge density increases when the segment is bent. The electrostatic contribution to ΔG will therefore be positive.

Turn now to equation (65), which gives a formula for $\Delta\bar{g}$, the free energy difference per mole phosphate, in units of RT , between two forms of DNA with charge densities ξ_{final} and ξ_{initial} . Identify the former with $\xi(\theta)$ and the latter with ξ . Expand $\Delta\bar{g}$ in powers of θ , noting that c_1^{lo} is a function of ξ through V_P as well as directly, and retain the lowest-order term only. The result is

$$\Delta\bar{g} = \Delta\bar{g}^0 + (\theta^2/24) \xi^{-1} [2\xi - 1 + \ln(c_1^{\text{lo}}/c_1) + \ln \kappa b], \quad (92)$$

where $\Delta\bar{g}^0$, the non-electrostatic contribution, is also assumed proportional to θ^2 . In this form it appears that the large positive contribution 2ξ is independent of the counterion diffusion potential. However, it should be remembered that equation (92) is simply a differential form of equation (65); the term 2ξ originates in the expansion of

$$(1 - \xi^{-1}) \ln(c_1^{\text{lo}}/c_1).$$

Substitution of (92) into (85) yields

$$a = a_0 + (l^2/12b) \xi^{-1} [2\xi - 1 + \ln(c_1^{\text{lo}}/c_1) + \ln \kappa b], \quad (93)$$

where a_0 is the contribution to the persistence length originating from the nonelectrostatic free energy $\Delta\bar{g}^0$ of bending. With the numerical values for duplex DNA in Table 1,

$$a = a_0 + 0.0815l^2 - 0.0134l^2 \log c_1, \quad (94)$$

where a and l are in ångströms.

Comparison of (94) with either of the empirical equations (83) or (84) shows identical forms if l is independent of ionic strength. If the slope of the $\log c_1$ dependence in (83) is compared with that of (94), the numerical value of l emerges as

$$l = 204 \text{ Å}. \quad (95)$$

If (84) is used instead, a similar value 194 Å is found for l . If (95) is now substituted back into (94), the latter becomes

$$a = a_0 + 3392 - 559 \log c_1. \quad (96)$$

The discussion to follow is based on (95) and (96), which in turn follow from comparison of (94) with (83); if (84) had been used instead, nothing substantive would require change.

C. Discussion of the numerical results

To assess the theory properly it is necessary to have some feeling for the numbers involved. The increment in b corresponding to a small bending angle θ , given by (90), is quite plausible. With $b = 1.7 \text{ \AA}$ and θ taken as one radian, Δb is about a tenth of an ångström, a value that could easily be the consequence of a slight structural distortion of the charge distribution. The value of about $200 \text{ \AA}$ (about 60 base pairs) for l is equally consistent. Recall that l is the length of a DNA segment which must retain rigid mechanical properties over the range of ionic strengths, $c_1 \lesssim 1 \text{ M}$, represented by the empirical (83) or (84). Reference to Fig. 7 shows that the persistence length is maintained at a value of about $450 \text{ \AA}$ even at very high ionic strengths. Therefore a segment l somewhat less than half this length does behave nearly like a rigid rod (Bloomfield *et al.* 1974).

At 0.1 M ionic strength (96) predicts that the polyelectrolyte contribution to the persistence length (sum of last two terms) is $3950 \text{ \AA}$, whereas the observed persistence length at 0.1 M from Fig. 7 is only $750 \text{ \AA}$. *The net effect of other contributions is to render DNA more flexible.* In more general quantitative terms, the value of a_0 calculated by comparison of (96) and (83) is negative,

$$a_0 = -3171 \text{ \AA}. \quad (97)$$

To see if this number is physically sensible, let us translate it into an equivalent free energy by applying (85) with ΔG and a replaced by ΔG_0 and a_0 , respectively. At 25°C , and with a_0 and l given by (97) and (95), respectively, ΔG^0 has the value -4.6 kcal for an angle θ of 1 radian. Bending of a segment of length l would be a spontaneous process if polyelectrolyte effects are ignored, driven by the free energy $-4.6 \text{ kcal}/(\text{radian})^2$. Since the segment l contains $(204/3.4) = 60$ base pairs, this free energy is a mere $-77 \text{ cal/mole base pair}$, and it is easy to imagine an increased entropy of base motion (increased breathing) when DNA is locally bent as a possible source of this free energy.

D. *Implications for DNA condensation and nucleosome formation*

According to the above analysis, removal of the polyelectrolyte effect, or counterion diffusion potential, at ionic strengths for which (83) and (96) are valid (a range that includes physiological conditions) would have as a consequence the spontaneous bending, or folding, of the DNA molecule. Mathematically, removal of the counterion diffusion potential is accomplished by setting the valence of a phosphate group equal to zero; the last two terms in (96) are then replaced by zero, and a reduces to the negative quantity a_0 . Physically, 'removal' of the polyelectrolyte effect at 0.1 M salt, for example, occurs at sufficiently high levels of multivalent cation binding. For example, even bound loosely, a polylysine molecule, or possibly two acting simultaneously, can completely neutralize the phosphate charge in its vicinity, neutralization being understood in the long-range sense of zero net charge. Or saturated binding of trivalent spermidine, which produces

$$1 - (3\xi)^{-1} = 92\%$$

phosphate charge neutralization, might be enough. The prediction is that DNA should then spontaneously fold, or 'condense', and that is exactly what happens (Gosule & Schellman, 1976). If the histone core of a nucleosome possesses a sufficiently high positive surface charge density, the possibility is also raised that a nucleosome is formed by the spontaneous wrapping of DNA around the protein core, driven solely by the non-specific counterion diffusion potential.

It is interesting that quite different considerations have led Riemer & Bloomfield (1978) to identify electrostatic repulsion as the primary resistance to DNA packaging in phage heads.

E. *Further discussion*

The procedural simplicity and appealing results obtained here should not be allowed to obscure a number of questions that can be raised. The extent to which the theoretical conclusions may depend on the accuracy of the data is not clear. Persistence length measurements are not straightforward. The dependence of Harrington's values on ionic strength is considerably steeper than that indicated by previous estimates (Harrington, 1978). His methods, however, do appear to be considerably more direct than those used previously. At any rate the experimental data require careful assessment, confirmation, and extension to include more data points at low ionic strength.

An apparent difficulty with the theoretical interpretation is illustrated by the behaviour of poly A, which is rigid when its bases are stacked at low temperatures (Stannard & Felsenfeld, 1975), but becomes flexible as the bases unstack in response to a temperature increase at a fixed ionic strength. The charge densities of unstacked and stacked single strands are similar (Olson & Manning, 1976). Stiffness, therefore, appears to be correlated with base stacking rather than with electrostatic repulsion. However, these two aspects of polynucleotide structure are not necessarily independent. A stacked conformation may provide the steric constraints that render obligatory the accompaniment of bending by the quasi-linear charge compression assumed in the theory; bending of an unstacked single strand may not require so severe an increase in electrostatic stress.

The behaviour of the persistence length at high ionic strengths, outside the range of validity of (96); is interesting and may also be a source of concern. An important distinction must be drawn here between the polyelectrolyte effect, or counterion diffusion potential, and electrostatic effects in general. By definition, mathematical reduction of the phosphate valence to zero abolishes all electrostatic effects along with the counterion diffusion potential. Sufficient cationic binding accomplishes the same thing. However, augmentation of the ionic strength to high levels may not. Although the counterion diffusion potential disappears at high ionic strength, nearest-neighbour electrostatic repulsion may survive because of the finite size of the screening ions. Indeed, the nearest-neighbour phosphate-phosphate distance of about 6–7 Å is not much larger, if at all, than current estimates of the effective hydrated diameter of the Na⁺ ion. Under these conditions (96) would be replaced by

$$a = a_0 + a_{nn,el}, \quad (98)$$

where a_0 would have about the same value as given by (97) and the nearest-neighbour electrostatic phosphate-phosphate repulsive contribution $a_{nn,el}$ is the only survival of the last two terms of (96). Both terms on the r.h.s. of (98) would be expected to depend slightly on ionic strength indirectly through solvent effects, but I shall assume ionic-strength independence as a first approximation. Fig. 7 shows that $a \approx 450$ Å at high ionic strength. Hence, with (97), an estimate for $a_{nn,el}$ is 3600 Å; translated into a free energy of bending by means of (85), this value is equivalent to a bending free energy of 5.2 kcal/(radian)² for the segment of length l , or only 87 cal/mole base pair/

(radian)². As a consistency check, I note that unscreened coulomb repulsions between each of the two phosphates in a base pair and their respective nearest neighbours at distance about 7 Å (which happens also to be about equal to the Bjerrum length) gives a free energy of bending of about $(2RT/24) = 50$ cal. This rough estimate assumes applicability of the bulk dielectric constant and totally unscreened phosphate charges; the factor $(24)^{-1}$ appears in (90).

Let us recapitulate. Even at high ionic strength segments of length $l \approx 200$ Å maintain their rigidity, since a is still observed to be more than twice as great. The rigidity under these conditions cannot be attributed to the counterion diffusion potential, which emerges only at lower ionic strength as the cause of increased segmental rigidity; a 200 Å segment becomes *harder* to bend as the ionic strength decreases, but it is still stiff even at high ionic strength, at least up to 2 M. The residual stiffness at high ionic strength can be attributed to surviving nearest-neighbour repulsions among the phosphates. If the latter could also be eliminated at high ionic strength, then DNA would become locally flexible in accord with what would then be the negative value a_0 of the persistence length. Recall again that at lower ionic strength, including physiological, *all* electrostatic effects can be eliminated by binding of multivalent cationic substances, leaving a_0 as the only contribution and thus rendering DNA locally flexible.

During the course of fifteen years of experience in polyelectrolyte theory I have accumulated many debts to many people. Special thanks are due to Bruno Zimm for initial motivation and guidance; to U. P. Strauss for sustained interest and encouragement; to Wilma Olson for whatever rudimentary knowledge of structure I may possess; to Alfred Holtzer, Tom Record and the late Lars Onsager for particular insights that led to new advances; and to Will McClure and Gary Felsenfeld for recent valuable discussions.

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